Retail Ebb and Flow and the Overnight–Intraday Return Gap*

Yongkil Ahn† Alfred Qi Fan‡ Dongwoo Noh§ Stella Yeayeun Park¶

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Abstract

In most stock markets, average overnight returns substantially exceed average intraday returns. Using investor-level flow data from Korea, we show that retail trading intensity drives this gap. We establish causality by instrumenting for retail trading proportion with nominal share prices, which affect retail trading through accessibility constraints but carry no fundamental information. We attribute this to "retail ebb and flow": driven by sensation-seeking and illusion of control, retail investors seek daytime exposure by systematically net buying at the open and net selling at the close. Consistent with this, day trading concentrates among young, male investors and intensifies following high volatility.

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†Seoul National University of Science and Technology (yongkil.ahn@seoultech.ac.kr)
‡HKUST (alfredfan@ust.hk)
§HKUST (noh@ust.hk)
¶Sogang University (yeayeunpark@sogang.ac.kr)

1 Introduction

A robust feature of global equity markets is the sharp contrast between returns realized during trading hours and those accrued overnight. Stocks earn substantial premiums from market close to the next day's open, while intraday returns are frequently negligible or even negative.[1] This overnight–intraday return gap persists across time and geographies, suggesting it stems from fundamental investor behavior interacting with market structure rather than being a spurious relationship. Recent literature attributes this pattern to investor heterogeneity. Distinct clienteles with differing objectives dominate trading at the open versus the close, and their systematic flows generate predictable price pressure (Lou et al. 2019). However, identifying which investors drive this wedge and why requires granular, high-frequency data on investor identities.

In this paper, we show that retail trading plays a central role in generating the overnight–intraday return gap. To do so, we use trade-level data from the Korea Exchange (KRX), which offers two advantages: (i) investor types, including retail investors, are identified for each trade directly by the KRX and (ii) the KRX is the sole stock exchange in Korea throughout our sample period. We thus work with accurate and exhaustive flows that can be safely called *the* retail flow, an advantage over U.S. studies where retail flows are either imputed or partial (Han and Kumar 2013, Lou et al. 2019, Boehmer et al. 2021, Barber et al. 2022a).

In both the time series and the cross section, retail trading proportion (RTP), the share of a stock's trading volume executed by retail investors, positively predicts the overnight–intraday return gap. When aggregate RTP is above its median in a month, the market return gap, the overnight return less the intraday return, averages 4.10% in the following month, twice the 2.05% that follows low-RTP months. In the cross section, stocks in the top RTP decile of a given month earn a 14.53% return gap in the following month, against 1.89% for the bottom decile. The cross-sectional relationship survives double sorts on size and volatility and persists in Fama–MacBeth regressions with conventional stock-level controls and industry fixed effects. The effect is also economically large. In the Fama–MacBeth regression with the full set of controls, a one-standard-deviation increase in RTP is associated with a 2.35pp higher return gap in the following month, 43% of the average stock-month gap of 5.52%.

We then ask whether RTP causes the gap rather than merely predicting it. We instrument RTP with

[1] See Figure A5 for global evidence. A notable exception is China, where a $T+1$ settlement rule prohibits same-day position reversals (Qiao and Dam 2020). The daytime-exposure mechanism we propose is consistent with this pattern.

the nominal share price, exploiting retail investors' tendency to trade low-priced stocks more actively (Du et al. 2022). The exclusion restriction is that, conditional on market capitalization and other firm characteristics, nominal price affects returns only through retail participation. This restriction is plausible because, at a given market capitalization, nominal price depends solely on share count, which does not itself affect firm fundamentals. Yet nominal price can affect trading accessibility: when fractional-share trading is unavailable, low-priced stocks are more accessible to investors with small account balances, who are disproportionately retail investors. Using account-level data from a national Korean brokerage firm, we find that, even after controlling for demographics and trading experience, a one-standard-deviation decrease in the logarithm of average daily account balance is associated with a 6.0pp increase in the proportion of volume executed in stocks with opening prices below KRW 10,000 (approximately USD 9).

The first stage is strong (Kleibergen–Paap F-statistic > 300), and the two-stage least squares (2SLS) estimate is 4.51pp per standard deviation of RTP, which is larger than the corresponding ordinary least squares (OLS) estimate. We attribute this to three factors. First, sophisticated investors can trade against predictable return patterns, which dampens the reduced-form relationship between lagged RTP and future return gaps. Second, the 2SLS coefficient is a local average treatment effect (LATE) that gives greater weight to stocks whose retail participation responds more strongly to nominal price. Our brokerage evidence suggests that this response operates through lower-balance investors, who favor low-priced stocks and also day-trade more. If their participation generates stronger price pressure, this weighting could explain the larger 2SLS estimate. Lastly, our regressions relate next month's returns to this month's RTP. Part of the month-to-month variation in RTP is transitory and does not carry into the next month, which attenuates the OLS estimate. The instrument picks up only the persistent, accessibility-driven component of RTP, so that the 2SLS estimate can exceed the OLS estimate (Appendix C). A remaining concern is that nominal prices may be related to omitted characteristics such as lottery-like features and volatility, or may reflect endogenous corporate decisions (Weld et al. 2009). However, insofar as these factors affect returns primarily by attracting retail investors, instrumenting RTP with nominal price remains appropriate.

We next document a stylized fact that explains how retail trading drives the return gap: retail investors, and mobile traders in particular, systematically net buy at the market open and net sell near the close, a pattern we refer to as "retail ebb and flow." High-RTP stocks have a larger share of investors who routinely open positions at the open and close them by day's end. RTP thus serves as a proxy for the prevalence of traders seeking daytime stock exposure, and high-RTP stocks exhibit

larger return gaps. Consistent with this interpretation, high RTP predicts stronger net buying near the open and stronger net selling near the close among investors who seek daytime exposure, namely mobile traders in the exchange data and day traders in the brokerage data. This ebb and flow responds to RTP as strongly over the following 5 and 20 days as on the next day. The opposite signs at the two ends of the day, and their persistence over the following weeks, distinguish our mechanism from attention-based explanations (Berkman et al. 2012). If retail investors simply concentrate demand following a salient event, we would observe positive flows at both the open and close, fading as attention subsides. Instead, we document renewed net buying at the subsequent open following net selling at the prior close.

To see whether these flows generate price pressure, we compare retail transaction prices to those of short-term institutions within the same stock, day, and 30-minute interval.[2] Relative to volume-weighted average prices (VWAPs) of the short-term institutions, mobile traders buy at prices 5.44bps higher in the first half hour of trading and sell at prices 4.50bps lower in the final interval before the close. Other retail traders exhibit a similar pattern that is weaker at the open (2.27bps) and comparable at the close (4.97bps). This behavior indicates aggressive, impatient trading consistent with price pressure from the retail ebb and flow.

Why do retail investors prefer such daytime exposure? We posit that this behavior reflects sensation-seeking (Grinblatt and Keloharju 2009) and overconfidence (Barber and Odean 2000, 2001a). Overconfidence in this context operates through the illusion of control (Barber and Odean 2001b). Retail investors derive utility from monitoring positions and managing them actively during market hours, believing such active engagement improves outcomes. They close them by day's end, as overnight positions offer neither monitoring opportunities nor the perception of control. In the same regressions that relate the ebb and flow to RTP, past idiosyncratic volatility likewise predicts stronger net buying at the open and stronger net selling at the close among mobile traders and day traders. This is consistent with our explanation, as sensation-seeking investors are drawn to volatility as a source of excitement (Grinblatt and Keloharju 2009) and volatile positions are the ones in which active monitoring seemingly matters most.

We further explore this interpretation in the brokerage data. First, we decompose trades originating from this brokerage into day trades, which are positions opened and closed within the same day, and non-day-trades. We find that the ebb-and-flow pattern is entirely driven by day

[2] Short-term institutions are financial investment companies, the category that includes most market makers (Section 2.2 and Appendix Table A7).

trades, while non-day-trade flows exhibit the opposite intraday pattern, net selling at the open and net buying at the close. Day trades account for 35–50% of volume in our brokerage sample, a surprisingly high share. Second, day trading propensity varies across trader characteristics in ways consistent with sensation-seeking and overconfidence. Regressing each investor's day-trade share of volume on demographics, we find that male traders engage in 2.94pp more day trading than female traders, younger traders 3.68pp more, and those with high self-reported investment knowledge 5.19pp more, after controlling for trading platform. Also, mobile and desktop app users engage in 6.06pp and 13.05pp more day trading relative to traders whose main channel is a branch office, a phone line, or the web. These effects are large against an average day trading proportion of 17.01%.[3] These effects compound, as young male traders reporting high investment knowledge day-trade the most. Day trading thus concentrates among the demographics that prior work associates with sensation-seeking and overconfidence.

We next relate our mechanism to alternative explanations of the return gap. In Fama–MacBeth regressions that progressively add controls for overnight systematic risk (Bogousslavsky 2021), attention-driven trading (Berkman et al. 2012), fast arbitrageur liquidity provision (Lu et al. 2023), and other variables proposed in prior studies, the coefficient on RTP remains robust and stable. We further test for heterogeneous effects by interacting RTP with indicators for high values of these other mechanism proxies. Limits to arbitrage suggest that these interactions should be positive, since arbitrageurs who find holding positions risky or costly are less willing to trade against the predictable price pressure created by retail investors' preference for daytime exposure (Shleifer and Vishny 1997, Pontiff 2006). The influence of retail trading is significantly amplified where idiosyncratic volatility is high, a proxy for heightened attention and tighter limits to arbitrage, and where retail order imbalance at the open is high, a proxy for costly liquidity provision. Our mechanism thus complements, rather than substitutes for, these explanations. The presence of arbitrageurs works in the opposite direction: where short-term institutions and foreign investors, the groups that take the other side of retail flow, trade more or absorb more of retail investors' morning buying, the RTP effect is about a quarter smaller (Appendix E).

Finally, the results are robust across samples. The RTP effect survives when we add the smaller half of stocks, split the sample by exchange (KOSPI and KOSDAQ) or by subperiod, and add cash dividends back to overnight returns.

[3]The cross-sectional average of trader-level day trading share is lower than the overall share because traders with the highest turnover ratios tend to be the most active day traders.

While our results come from Korea, a particularly retail-heavy market, we see its experience as a harbinger of retail trading trends around the world. Korea's market-level RTP has hovered around 70% over our sample period, well above traditional U.S. estimates of 5–20%. Recent data, however, show the U.S. retail share rising rapidly, with some sources estimating 30–37% of daily trading volume.[4] Mobile trading in Korea grew from under 4% of retail volume in 2010 to over half by 2020, a shift now under way in the U.S. and other major markets. Our message, that retail trading shapes return dynamics, is therefore likely to apply well beyond Korea in the years ahead.

Related literature. The overnight–intraday return gap is well-documented across multiple markets (Kelly and Clark 2011, Branch and Ma 2015). The literature most closely related to our work attributes this pattern to price pressure from distinct investor groups. Specifically, Lou et al. (2019, 2026) propose a "tug-of-war" between overnight and intraday clienteles. Berkman et al. (2012) link high overnight returns to retail attention, arguing that individual investors concentrate buy orders at the market open. While our framework aligns with this clientele-based perspective, we offer two refinements. First, our mechanism explains the *recurrent* retail ebb and flow driven by a systematic preference for daytime exposure. Second, we establish a causal link between retail intensity and the return gap.

Other work extends the tug-of-war framework to the predictability of returns (Akbas et al. 2022), to spillovers across firms (Wang 2025), and to end-of-day reversals (Baltussen et al. 2024). A second set of studies attributes the gap to risk and frictions, namely distinct day and night risk premia (Hendershott et al. 2020), inventory risk and arbitrage constraints (Bogousslavsky 2021, Boyarchenko et al. 2023), and cream-skimming among liquidity providers at the open (Lu et al. 2023). Others point to uncertainty resolution (Hu et al. 2022, Bondarenko and Muravyev 2023), extrapolation (Jones et al. 2022), sentiment (Aboody et al. 2018), the intraday origin of momentum (Barardehi et al. 2023), and the same asymmetry in option returns (Muravyev and Ni 2020). We treat these channels as complementary rather than competing, and Section 4.3 shows that the RTP effect survives controls for them and is amplified where attention and limits to arbitrage are strong.

Our work also belongs to the literature on retail investor behavior and its price impact (Barber et al. 2009). Previous studies have shown that retail investors are overconfident (Barber and Odean 2000, 2001a), attention-driven (Barber and Odean 2008, Barber et al. 2022b), and prone to speculation (Kumar and Lee 2006, Kumar 2009, Han and Kumar 2013), and they concentrate

[4] Boehmer et al. (2021) estimate U.S. retail share at 17.5% as of 2022, down from a COVID-era peak of 24%. More recent estimates from MEMX (2025) suggest retail has since rebounded to 30–37% of daily U.S. equity volume.

in small, illiquid, and volatile stocks (Laarits and Sammon 2025). Our paper shows that these predispositions can give rise to a preference for daytime exposure and hence to a recurring intraday pattern in retail flows. With respect to day trading, prior work has viewed it as investor learning (Linnainmaa 2011), skill (Barber et al. 2014), or noise (Dalvi et al. 2023), while Grinblatt and Keloharju (2009) and Dorn and Sengmueller (2009) tie it to sensation-seeking and entertainment. Our evidence favors the latter reading. Finally, we establish these results with flows in which retail identity is observed rather than inferred, complementing U.S. evidence built on increasingly precise imputation methods (Boehmer et al. 2021, Barber et al. 2022a).

Lastly, our results contribute to the literature on price pressure and limits to arbitrage (Coval and Stafford 2007, Kaniel et al. 2008). We show that predictable, high-frequency flows generate persistent mispricing that sophisticated investors do not fully offset. These findings underscore the presence of limits to arbitrage even at intraday horizons (Heston et al. 2010).

2 Data and stylized facts

We first describe the features of the KRX's market microstructure that matter for our analysis. We then describe our primary data sources—trade-level data from the KRX and account-level data from a large Korean discount brokerage—and document three stylized facts that motivate our analysis. First, overnight returns substantially exceed intraday returns in the Korean market. Second, retail investors are net buyers in the morning and net sellers in the afternoon, while institutions exhibit the opposite pattern. Third, a large fraction of retail trading volume is driven by day trades, defined as positions opened and closed within the same day.

2.1 Market microstructure of the KRX

The KRX is the sole stock exchange in Korea throughout our sample period. All orders therefore meet in a single limit order book, unlike in the United States, where trading fragments across exchanges, alternative trading systems, and off-exchange internalizers. Each trading day begins with an opening auction, is followed by a continuous trading session from 9:00 a.m. to 3:30 p.m. (3:00 p.m. before August 2016), and ends with a closing auction.[5] Both the opening and closing auctions are single-price call auctions in which the clearing price is chosen to maximize the volume matched

[5] Pre- and post-market sessions allow additional trading at the most recent daily closing price. Because these sessions are thinly traded, we exclude them from our intraday measures.

among orders in the book at the time of clearing. The opening auction accumulates orders during the half hour preceding 9:00 a.m. (a full hour before April 2019), whereas the closing auction accumulates orders during a ten-minute pause in continuous trading before the close. We compute overnight and intraday returns from the two auction-clearing prices. Appendix B examines the two auctions in detail and shows that our results are unchanged when returns are instead computed from VWAPs over the first and last thirty minutes of trading (Appendix Table A2).[6]

Several frictions raise the cost of short-horizon institutional strategies. A securities transaction tax of 0.30% of the sale value applies to every sale for most of our sample period, which renders thin-margin strategies such as high-frequency trading infeasible.[7] Short selling is restricted to covered positions and subject to an uptick rule, and it was banned outright for extended stretches of our sample, namely market-wide in 2011 and 2020–2021 and for all but the largest stocks thereafter. These frictions help explain why the predictable price pressure we document is not fully arbitraged away. Because they apply nearly uniformly across stocks and change slowly over time, however, they cannot explain why the gap arises or why it varies so strongly across stocks and over time.

2.2 KRX data

We obtain trade-level data for all common stocks listed on the KRX from January 2009 to December 2023. The KRX operates two major markets, the KOSPI (the main board) and the KOSDAQ (analogous to the NASDAQ). As of December 2023, these markets collectively hosted 2,541 listed common stocks.

Given that the average Korean listed firm is small relative to those listed on U.S. exchanges, we restrict our sample to the largest half of stocks ranked by beginning-of-year market capitalization to ensure that micro-cap stocks do not influence our results.[8] This filter yields an average of 1,015 stocks per year and a total of 169,961 stock-month observations over the sample period, accounting for an average of 97% of total market capitalization.

We obtain book equity, earnings, dividends per share, shares outstanding, and foreign ownership from the KRX's public data portal.[9] We compute market capitalization, return volatility, the Amihud (2002) illiquidity measure, and market beta from the trade-level data. Appendix A provides detailed

[6] All KRX-provided prices are adjusted for splits, reverse splits, and other capital changes, but not for cash dividends. Our results are robust to adding back cash dividends to the ex-date overnight return (Appendix Table A6).

[7] The rate was 0.30% until mid-2019 and was reduced in steps to 0.25% in 2019, 0.23% in 2021, and 0.20% in 2023, the final year of our sample.

[8] Appendix Table A6 confirms our main results in the full universe of stocks, including the smaller half.

[9] Foreign ownership is reported for regulatory reasons, but ownership shares of other investor types are not reported.

variable definitions.

Overnight and intraday returns. Our main outcome variables are overnight and intraday returns. We follow Lou et al. (2019) in decomposing the daily close-to-close return into overnight and intraday components. We define the intraday return as the open-to-close return:

$$r_{i,s}^{\text{intraday}} = \frac{P_{i,s}^{\text{close}}}{P_{i,s}^{\text{open}}} - 1,$$

where $P_{i,s}^{\text{open}}$ and $P_{i,s}^{\text{close}}$ denote the opening and closing prices of stock i on day s. We then define the overnight return as:

$$r_{i,s}^{\text{overnight}} = \frac{1 + r_{i,s}^{\text{close-to-close}}}{1 + r_{i,s}^{\text{intraday}}} - 1,$$

where $r_{i,s}^{\text{close-to-close}}$ is the full daily return of stock i on day s. Figure 1 illustrates this decomposition.

[Figure 1 here]

Figure 2 plots market capitalization-weighted cumulative overnight, intraday, and close-to-close log returns for all common stocks listed on the KRX. Overnight returns substantially exceed close-to-close returns, while intraday returns are negative on average, consistent with U.S. evidence in Berkman et al. (2012). Over our 15-year sample from the beginning of 2009 to the end of 2023, a $1 investment in the overnight component grows to approximately $21.84, while the same investment in the intraday component declines to roughly $0.12.

[Figure 2 here]

We define the overnight–intraday return gap at the daily level as the difference between the overnight and intraday returns:

$$\text{RG}_{i,s}^{d} = r_{i,s}^{\text{overnight}} - r_{i,s}^{\text{intraday}}.$$

For our main analysis, we aggregate to the stock-month level. Following Lou et al. (2019), we

compound daily returns within each month and define the monthly return gap as:

$$r_{i,t}^{\text{overnight}} = \prod_{s \in \text{month } t} \left(1 + r_{i,s}^{\text{overnight}}\right) - 1,$$
$$r_{i,t}^{\text{intraday}} = \prod_{s \in \text{month } t} \left(1 + r_{i,s}^{\text{intraday}}\right) - 1,$$
$$\text{RG}_{i,t}^{m} = \frac{1 + r_{i,t}^{\text{overnight}}}{1 + r_{i,t}^{\text{intraday}}} - 1.$$

The daily gap is a simple difference, whereas the monthly gap is a ratio of the compounded overnight and intraday legs. We refer to either as the return gap when the context is unambiguous.

Investor type-level trade flows. At the KRX, each executed trade is tagged with the investor type of both the buyer and the seller. KRX classifies investors into nine categories: (i) financial investment companies, (ii) insurance companies, (iii) investment trust companies, (iv) private equity funds, (v) banks, (vi) other financial companies, (vii) government pension funds, (viii) other corporations, and (ix) individuals (retail investors). We label financial investment companies as short-term institutions and group all other non-retail institutions as long-term institutions.[10]

In addition to investor type, KRX tags each trade with the trading medium through which it was executed: (i) brokerage, (ii) landline, (iii) mobile trading system (MTS), (iv) home trading system (HTS), or (v) direct market access (DMA). MTS and HTS are the brokerages' proprietary trading applications for smartphones and for desktop computers, respectively. We group brokerage, landline, and DMA into "other media" because, with the rise of mobile and desktop trading, they collectively account for less than 10% of retail volume since 2016. This classification yields five investor categories: short-term institutions, long-term institutions, MTS retail traders, HTS retail traders, and other retail traders.

The KRX also classifies each order as a limit or market order. However, this classification sometimes represents the demand for immediacy inaccurately, because limit orders priced to cross the book immediately are still recorded as limit orders. When we need the initiating side of a trade (Section 4.3 and Appendix E), we identify it from order arrival times: because all orders meet in a single limit order book, the later-arriving order is necessarily the taker order, and we treat it as the aggressive one.

[10]The KRX investor code for financial investment companies is 1000. We make this grouping because some institutional types, such as private equity and insurance companies, trade infrequently at short horizons. Appendix Table A7 supports this grouping. See the KRX data portal for additional details.

We document intraday trading patterns using 30-minute intervals. For each stock i, investor type j, and day s, we aggregate trades into 30-minute intervals from 9:00 a.m. to 3:00 p.m. Because trading hours were extended in August 2016 (Section 2.1), we fold all trades after 2:30 p.m., including the closing auction, into a single final interval, so that the last interval runs to the close in both regimes. The first interval likewise includes the opening auction. This yields 12 intraday intervals per day, indexed by $\tau \in \{1, \ldots, 12\}$. For each interval, we define:

$$\text{Buy}_{i,j,s,\tau} = \text{shares of stock } i \text{ bought by investor type } j \text{ during interval } (s, \tau),$$
$$\text{Sell}_{i,j,s,\tau} = \text{shares of stock } i \text{ sold by investor type } j \text{ during interval } (s, \tau).$$

We define net buy as buy minus sell, normalized by shares outstanding:

$$\text{Net buy}_{i,j,s,\tau} = \frac{\text{Buy}_{i,j,s,\tau} - \text{Sell}_{i,j,s,\tau}}{\text{Shares outstanding}_{i,s}}.$$

Normalizing by shares outstanding allows us to compare net buy across stocks and over time. We express net buy in basis points because trade flows are small relative to shares outstanding at the 30-minute frequency.

We also define the order imbalance (OIB) of an investor type as net buy scaled by the type's own trading volume during the same stock-day-interval

$$\text{OIB}_{i,j,s,\tau} = \frac{\text{Buy}_{i,j,s,\tau} - \text{Sell}_{i,j,s,\tau}}{\text{Buy}_{i,j,s,\tau} + \text{Sell}_{i,j,s,\tau}}.$$

Because OIB is bounded between -1 and 1, it captures the direction of a group's trading irrespective of the group's size, whereas net buy scaled by shares outstanding also reflects the size of the group's flow. Both normalizations are standard in the literature on investor order flow (e.g., Kaniel et al. 2008, Kelley and Tetlock 2013, Boehmer et al. 2021).

Figure 3 plots average net buy and OIB across stocks and days for each 30-minute interval, separately for retail investors, short-term institutions, and long-term institutions. Panel (a) shows that retail investors are net buyers in the morning (peaking in the first 30 minutes) and net sellers in the afternoon (especially in the final interval). Short-term and long-term institutions exhibit the opposite pattern, consistent with market clearing. In Panel (b), OIB shows a similar pattern, indicating that the intraday flow pattern is not an artifact of scaling by shares outstanding.

[Figure 3 here]

Panels (c) and (d) decompose retail flows by trading medium in the two normalizations. The morning-buying and afternoon-selling pattern is most pronounced among MTS users, suggesting that mobile traders drive the aggregate retail pattern.

An important independent variable in our analysis is RTP.[11] Han and Kumar (2013) impute RTP from small trade sizes in U.S. data, but investor type identifiers allow us to compute it directly in this study. At the stock-month level, we define RTP as:

$$\text{RTP}_{i,t} = \frac{\text{Retail volume}_{i,t}}{\text{Total volume}_{i,t}}.$$

Because both sides of every trade are tagged, the non-retail share of volume is simply 1 – RTP. While our main analysis focuses on RTP, Appendix E explores how the presence of different types of non-retail investors affects the return gap.

Summary statistics. Table 1 presents summary statistics for our main variables. The average stock-month overnight return is 2.37%, the average intraday return is −1.45%, and the mean return gap is 5.52%.[12] These magnitudes are comparable to U.S. evidence: Berkman et al. (2012) report daily overnight and intraday returns of 0.098% and −0.066%, which compound to approximately 1.96% and −1.32% per month.

[Table 1 here]

The median market capitalization in our sample is KRW 264 billion (approximately USD 233 million), reflecting the relatively small size of Korean listed firms.[13] The average RTP is 72.98%, well above traditional U.S. estimates of 5–20%. Korea's high retail intensity reflects the institutional frictions described in Section 2.1, together with the early and widespread adoption of mobile and desktop trading platforms.

Table 2 reports time-averaged cross-sectional correlations among our main variables. Monthly RTP has a positive correlation of 0.19 with the return gap, consistent with our hypothesis that retail trading drives the overnight–intraday return spread. However, RTP is strongly correlated

[11] We use RTP rather than retail net buy near the open minus retail net buy near the close (RNBO−RNBC) because market clearing mechanically links retail and institutional flows and can confound such measures. In particular, a high retail net buy can reflect not only strong retail buying but also weak or negative net buying by institutions (Appendix Figure A6).

[12] The mean return gap exceeds the difference between the two mean legs because the monthly gap is a ratio of the compounded legs rather than their difference.

[13] To facilitate interpretation, we convert KRW values to USD using the sample period average exchange rate of 1,130 KRW per USD.

with other firm characteristics. For example, high-RTP stocks tend to be small ($\rho = -0.74$ with log market equity), illiquid ($\rho = 0.11$ with Amihud illiquidity), and volatile ($\rho = 0.31$ with idiosyncratic volatility), consistent with Laarits and Sammon (2025). This motivates our use of control variables and instrumental variables in Section 3.

[Table 2 here]

Figure 4 shows that the positive relationship between RTP and the return gap persists within size terciles. We assign stocks to terciles based on average daily market capitalization each month, then create binscatter plots using 20 bins within each tercile. The monotonic relationship between RTP and the return gap appears in all three terciles, indicating that size alone does not explain this pattern.

[Figure 4 here]

2.3 Brokerage data

Our brokerage data are analogous to the U.S. discount-brokerage datasets in Barber and Odean (2000, 2001a) and the Chinese brokerage data in Feng and Seasholes (2005).[14] We observe anonymized account-level trades, end-of-day positions, and investor demographics. Our sample covers approximately 235,000 active retail accounts at a large Korean discount broker from January 2015 to December 2018. We receive data on all traders who execute at least 10 trades during the sample period, which represents approximately 7.8% of total retail trading volume during the same period.

The trade file reports all executions (quantity, price, timestamp, and order medium), the balance file contains end-of-day holdings for all securities, and the demographic file includes sex, birth date, account opening date, and other self-reported information. Korean brokerages collect self-reported information on income, investment knowledge, investment aggressiveness, and derivative experience to determine investor suitability for different investment products, as required by Korean financial regulations. The specific questions asked are included in Panel F of Appendix Table A1.

Using the trade file, we identify round-trip positions that qualify as day trades. For a given investor, stock, and day, we classify the investor's trading in that stock as day trading if the investor both buys and sells the stock during the day, that is, if a position opened during the day is fully or

[14] The brokerage firm's identity cannot be disclosed under the terms of our data-use agreements.

partially closed during the same day.[15] All trade volume is thus partitioned into day-trade volume, the volume in such stock-days, and non-day-trade volume, the rest. For each day, we compute the fraction of total executed volume attributable to day trades. Figure 5 plots the monthly time series of the day-trade volume share. Computing this quantity requires account-level data to track position-level round trips, which explains why day trade volume share is rarely discussed in the literature. The proportion is surprisingly high, averaging around 45% over the sample period.

[Figure 5 here]

To examine whether day traders drive the intraday flow pattern documented in Figure 3, we decompose retail net buy into day-trade and non-day-trade components. Figure 6 shows the day-trade and non-day-trade flows alongside the total brokerage retail flow. The morning-net-buying and afternoon-net-selling pattern is entirely driven by day-trade flows. This suggests that the retail ebb and flow we document largely reflects the behavior of investors who open and close positions within the same day.[16]

[Figure 6 here]

The brokerage data also allow us to distinguish traders by interface. We classify traders as MTS users, HTS users, or others based on the order medium recorded at execution. For each investor, we define the main medium as the interface through which she executes the greatest number of trades. We use these categories to characterize differences in activity, day-trade intensity, and performance.

Table 3 presents summary statistics for our brokerage sample. Panel A shows that average account balances and trade sizes are comparable to those in Seru et al. (2010) who use a Finnish dataset. The age and gender composition are also similar, with a mean user age of 46.15 and 61% of users being male.

[Table 3 here]

Panel B compares users across trading platforms. MTS users are the youngest (mean age 42) and hold the smallest account balances (mean KRW 23.26 million), while also reporting the lowest

[15] Korean regulation places no restriction on day trading during our sample period. There is no analogue of the U.S. "pattern day trader" rule (FINRA Rule 4210), which required margin accounts executing four or more day trades within five business days to maintain a minimum equity of USD 25,000.

[16] Net buy scaled by shares outstanding diverges in level across Figures 3 and 6, because the brokerage accounts for roughly 8% of retail volume and under-weights mobile trading, 25% of its trading value against 37% on the exchange over 2015–2018.

levels of income and investment experience. In contrast, HTS users are older (mean age 50) and manage the largest balances (mean KRW 68.68 million). Users of other traditional channels (labeled as "Other"), such as brokerage-assisted or web-based trading, are the oldest group (mean age 52) but are less active traders.

Both MTS and HTS users engage in substantially more day trading than traditional media users. Day trades make up 17.11% of the average MTS user's volume and 24.11% of the average HTS user's, compared to only 10.31% for users of other media. This pattern suggests that technology-enabled trading platforms, whether mobile or desktop, facilitate intraday trading. In addition, MTS users exhibit signs of lower sophistication. They self-report the lowest investment knowledge, with a mean of 2.36 against 2.76 for HTS and 2.64 for traditional media on a scale from 1 (little understanding) to 4 (understanding of most products including derivatives). They also realize the worst performance, with a mean return of −4.75% compared to −3.13% for HTS and −3.44% for traditional users.[17]

3 Empirical results

We begin by documenting time-series variation in the overnight–intraday return gap at the market level and find that aggregate RTP is positively associated with the gap. We then turn to cross-sectional analysis and find that stocks with higher RTP earn higher overnight returns and lower intraday returns. Finally, we establish a causal link from RTP to the return gap using an instrumental variables approach that exploits variation in nominal stock prices.

3.1 Time-series evidence

Each day, we compute value-weighted average overnight and intraday returns, weighting by beginning-of-day market capitalization, and compound the daily series within each month for the monthly analysis. We compute aggregate RTP in period t as the ratio of retail trading volume to total trading volume (both measured in KRW) across all stocks in our sample. We then split the sample into high-RTP and low-RTP periods based on whether period t market-level RTP is above or below its time-series median, and examine returns in period $t+1$.

Table 4 reports average overnight returns, intraday returns, and the return gap following high-RTP and low-RTP periods. Panel A splits the sample by whether daily (monthly) RTP in the preceding period is above or below its median. At the daily level, the overnight return averages 0.11% following

[17] See Appendix Table A1 for how we compute realized returns.

high-RTP days versus 0.06% following low-RTP days, a difference of 0.05pp (significant at the 5% level). The return gap is 0.18% following high-RTP days versus 0.10% following low-RTP days, a difference of 0.08pp (significant at the 5% level). Intraday returns are slightly more negative following high-RTP days (−0.06% versus −0.04%), though the difference is not statistically significant.

[Table 4 here]

These patterns are more pronounced at the monthly level. Overnight returns average 2.40% following high-RTP months versus 1.20% following low-RTP months, a difference of 1.19pp (significant at the 10% level), and return gaps average 4.10% versus 2.05%, a difference of 2.06pp (significant at the 1% level). Intraday returns are −1.56% following high-RTP months versus −0.76% following low-RTP months, a difference of −0.80pp (significant at the 10% level).

Panel B presents results from predictive regressions where we regress next period returns on current-period aggregate RTP, denoted as RTP_t. Specifically, we estimate the following regression:

$$y_{t+1} = \alpha + \beta \text{RTP}_t + \gamma r_t + \delta \sigma_t + \varepsilon_{t+1} \tag{1}$$

where y_{t+1} may be overnight return, intraday return, and return gap of aggregate market in period $t+1$, r_t is the close-to-close market return, and σ_t is market volatility. In daily regressions, σ_t is the standard deviation of daily market returns over the preceding 20 days. The monthly analogue is the same 20-day standard deviation measured on the last trading day of month t.

At the daily level, a 1pp increase in RTP is associated with a 0.39bps increase in next-day overnight returns (significant at the 5% level) and a 0.70bps increase in the return gap (significant at the 1% level). At the monthly level, the predictive power is stronger. A 1pp increase in RTP is associated with a 9.05bps increase in next-month overnight returns (significant at the 5% level) and an 18.07bps increase in the return gap (significant at the 1% level).

These results show that periods with more intense retail trading are followed by higher overnight returns, lower (more negative) intraday returns, and a larger overnight–intraday return gap.

3.2 Cross-sectional return predictability

In the cross section, we find a positive relationship between RTP and the subsequent return gap in both portfolio sorts and Fama–MacBeth regressions.

Portfolio sorts. Each month, we sort stocks into deciles based on RTP during month t and compute value-weighted portfolio returns over month $t+1$. Table 5 reports average monthly overnight returns, intraday returns, and the return gap for each decile portfolio.

[Table 5 here]

Panel A shows that overnight returns increase almost monotonically from 0.89% in the lowest RTP decile to 5.27% in the highest RTP decile, a spread of 4.37pp. The alpha from the Fama–French five-factor model estimated for the Korean market (α_{FF5}) also increases, from 0.70% to 4.94%, with a high-minus-low alpha of 4.24pp.[18] Panel B shows that intraday returns decline from −0.30% in the lowest RTP decile to −5.47% in the highest RTP decile, a spread of −5.16pp. The α_{FF5} spread is −4.92pp. Panel C reports the monthly return gap RG^m. The return gap increases monotonically from 1.89% in the lowest RTP decile to 14.53% in the highest RTP decile, a spread of 12.64pp. The α_{FF5} spread is 12.29pp. All t-statistics are computed using Newey–West standard errors with six lags to account for potential autocorrelation in monthly returns. These large, statistically significant spreads show that RTP reliably predicts the overnight–intraday return gap, with nearly monotonic patterns across deciles for overnight and intraday returns and a monotonic pattern for the gap.

One concern is that RTP is correlated with firm size and return volatility (Table 2). To address this, we perform dependent double sorts on RTP and size, and on RTP and volatility, sorting first by size (volatility) and then by RTP within each size (volatility) quintile. Table 6 reports the results. Panel A shows that the return gap monotonically increases with RTP within each size quintile. For example, among small stocks, the return gap increases from 2.56% in the low-RTP quintile to 9.24% in the high-RTP quintile, a spread of 6.68pp. Among large stocks, the spread is 8.23pp. Panel B shows similar results for double sorts on RTP and volatility. The return gap spread is 2.82pp among low-volatility stocks and 11.54pp among high-volatility stocks. These results confirm that the predictive power of RTP is not merely driven by its correlation with size or volatility. However, the RTP effect is notably stronger among high-volatility stocks, consistent with the view that overnight risk makes it more difficult to arbitrage away the price pressure created by the retail clientele.

[Table 6 here]

Fama-MacBeth regressions. To control for multiple firm characteristics simultaneously, we estimate Fama–MacBeth regressions. Specifically, we estimate the following cross-sectional regression

[18] The Korean factors are obtained from Global Factor Data by Jensen et al. (2023).

each month:

$$y_{i,t+1} = \alpha_t + \beta_t \text{RTP}_{i,t} + \mathbf{X}'_{i,t}\Gamma_t + \varepsilon_{i,t+1}, \tag{2}$$

where $y_{i,t+1}$ is the overnight return, intraday return, or return gap for stock i in month $t+1$, and $\text{RTP}_{i,t}$ is RTP in month t. The vector $\mathbf{X}_{i,t}$ includes size, book-to-market, market beta, Amihud illiquidity, foreign ownership, momentum, idiosyncratic volatility, and maximum daily return. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally each month to have zero mean and unit standard deviation. We report time-series averages of the coefficients β_t and Γ_t with Newey–West standard errors using six lags. Appendix Table A1 provides detailed variable definitions.

Table 7 reports the results. In columns 1, 4, and 7, a one-standard-deviation increase in RTP is associated with a 1.24pp increase in next-month overnight returns, a 1.22pp decrease in intraday returns, and a 2.97pp increase in the return gap, all significant at the 1% level.

[Table 7 here]

Columns 2, 5, and 8 add current overnight and intraday returns, size, market beta, book-to-market, Amihud illiquidity, foreign ownership, momentum, idiosyncratic volatility, and maximum daily return as controls. The coefficient on RTP remains statistically significant across all three outcome variables, though the magnitudes are somewhat smaller than in the univariate specification, indicating that part of RTP's predictive power operates through correlated firm characteristics.

Columns 3, 6, and 9 add industry fixed effects based on KRX classifications. Even with controls and industry fixed effects, the effect remains economically large. A one-standard-deviation increase in RTP is associated with a 2.35pp increase in the return gap, or 43% of the unconditional mean return gap of 5.52% (Table 1). The adjusted R^2 values indicate that RTP and controls explain substantial cross-sectional variation in returns: 4.37% to 15.69% for overnight returns, and 4.11% to 16.02% for the return gap. These magnitudes are comparable to those in Lou et al. (2019).

The controls reveal several additional patterns. Consistent with Berkman et al. (2012), idiosyncratic volatility is positively associated with the return gap, while past overnight returns generate short-term continuation. Interestingly, the coefficient on size is positive, seemingly at odds with Berkman et al. (2012), who find smaller stocks have higher overnight returns. This reversal occurs because RTP is negatively correlated with size ($\rho = -0.74$, Table 2). Within each size tercile, the return gap increases with RTP, but smaller stocks have higher RTP on average (Figure 4). Once we control for RTP, the residual size effect is positive.

3.3 Instrumental variables approach

While the Fama–MacBeth regressions establish a strong association between RTP and the return gap, the relationship may still be subject to omitted variables bias or reverse causality. Sophisticated investors may trade against predictable return gaps, attenuating OLS estimates, or unobserved firm characteristics may simultaneously attract retail trading and generate return gaps through independent channels.

To establish a causal link, we use an instrumental variables approach that exploits retail investors' preference for low-priced stocks (Birru and Wang 2016, Du et al. 2022). We verify this pattern in our brokerage data and use log nominal share price on the first trading day of each month as our instrument. The per-share price of a stock should carry no economic meaning. Conditional on market capitalization, a $100 stock with 1 million shares outstanding is economically identical to a $10 stock with 10 million shares. However, in the absence of fractional share trading, which was not readily available to Korean retail investors during our sample period, stocks with lower per-share prices are more accessible to investors with small account balances.[19]

Accessibility constraints. We first validate this mechanism using account-level data from our brokerage sample. For each investor, we compute the average daily account balance over the sample period and the proportion of trading volume (in KRW) executed in stocks with opening prices below three thresholds: KRW 1,000, KRW 5,000, and KRW 10,000. We also compute the volume-weighted average opening price across all stocks traded by each investor. We then residualize both variables with respect to investor demographics (sex, age, income, investment knowledge, derivative experience, and trading medium) to isolate variation in trading behavior that is not explained by observable characteristics.

Figure 7 plots binned scatter plots of these residualized variables against residualized log average daily balance. Panel (a) shows that investors with smaller account balances concentrate their trading in low-price stocks. A one-standard-deviation decrease in the log of average daily account balance is associated with a 6.0pp increase in the proportion of volume executed in stocks with opening prices below KRW 10,000 (significant at the 1% level). Panel (b) shows that investors with smaller balances trade stocks with lower prices. A 10% decrease in average daily balance is associated

[19] Fractional trading of domestically listed shares became possible only in September 2022, near the end of our sample, through a beneficiary-certificate scheme approved by the Financial Services Commission. Fractional trading of foreign stocks had been permitted since 2019.

with a 1.8% decrease in the volume-weighted average opening price. These patterns confirm that accessibility constraints drive retail investors' preference for low-priced stocks.

[Figure 7 here]

2SLS regressions. We estimate the following 2SLS specification:

$$\text{RTP}_{i,t} = \alpha_{\text{FS}} + \gamma \log(\text{Nominal price})_{i,t} + \mathbf{X}'_{i,t}\Gamma_{\text{FS}} + \text{Month FE} + \text{Industry FE} + \varepsilon_{i,t}, \tag{3}$$

$$y_{i,t+1} = \alpha + \beta\widehat{\text{RTP}}_{i,t} + \mathbf{X}'_{i,t}\Gamma + \text{Month FE} + \text{Industry FE} + \eta_{i,t+1}, \tag{4}$$

where $y_{i,t+1}$ is the overnight return, intraday return, or return gap for stock i in month $t+1$, Nominal price$_{i,t}$ is the closing share price on the first trading day of month t, unadjusted for splits, $\mathbf{X}_{i,t}$ includes the same controls as in Table 7, and $\widehat{\text{RTP}}_{i,t}$ is the fitted value from the first stage. We include month and industry fixed effects, based on KRX classifications, and cluster standard errors by stock and month.

The exclusion restriction requires that, conditional on market capitalization and firm characteristics, nominal price affects returns only through its effect on RTP. This is plausible since nominal price carries no information about fundamentals or cash flows, and any effects on investor attention or trading behavior should operate through retail trading activity, which is captured by RTP. The first stage is strong, with a Kleibergen–Paap (KP) rank Wald F-statistic of 318.43 (Table 8), well above conventional thresholds for weak instruments (Stock and Yogo 2005, Kleibergen and Paap 2006).[20]

Table 8 reports the results. The OLS coefficient on RTP is 2.36, indicating that a one-standard-deviation increase in RTP is associated with a 2.36pp increase in the return gap. The 2SLS coefficient is 4.51, larger in magnitude than the OLS estimate.

[Table 8 here]

Three factors help explain the difference. First, as noted above, sophisticated investors can trade against predictable price pressure, which compresses the observed return gap and attenuates the reduced-form OLS estimate. Second, the 2SLS coefficient is a LATE, which gives greater weight to stocks whose retail participation responds more strongly to nominal price. The brokerage evidence in Figure 7 suggests that this response operates through lower-balance investors, who favor low-priced

[20] Alternative instruments built from nominal price, namely a quadratic in log price, a piecewise spline over tick brackets, and nominal-price-bin indicators, deliver weaker first stages. Their second-stage estimates for the return gap lie between 4.37 and 4.58, and we therefore retain the parsimonious log-price instrument (Appendix Table A5).

stocks and also day-trade more.[21] The estimate may therefore place greater weight on stocks whose retail clientele has a stronger preference for daytime exposure. If participation by these investors generates greater price pressure, this weighting could help explain the larger 2SLS estimate. Third, to the extent that RTP contains a sizable transitory component, OLS with lagged RTP is attenuated because it loads on variation that does not persist into the next period, whereas the instrument shifts a more persistent, accessibility-driven component of retail participation.[22]

4 Mechanism

We now explore why higher RTP causes a larger overnight–intraday return gap. First, we establish that high-RTP stocks exhibit stronger retail ebb and flow in subsequent periods, with larger net buying near the open and larger net selling near the close among mobile traders and day traders. Second, we examine why retail investors prefer daytime exposure by relating day trading behavior to trader characteristics. Third, we assess whether our mechanism is robust to controls for alternative explanations and test whether it operates as a substitute or complement to other proposed drivers of the return gap.

4.1 RTP predicts retail ebb and flow

We test whether RTP predicts retail net buying near the open and net selling near the close by estimating the following Fama–MacBeth regressions:

$$\text{Flow}_{i,t+\tau} = \alpha_t + \beta_t \text{RTP}_{i,t} + \mathbf{X}'_{i,t}\Gamma_t + \varepsilon_{i,t+\tau}, \tag{5}$$

where $\text{Flow}_{i,t+\tau}$ is retail net buy, mobile trader net buy, or day-trade net buy at the open or close for stock i, averaged over the subsequent $\tau \in \{1, 5, 20\}$ days, $\text{RTP}_{i,t}$ is RTP on day t, and $\mathbf{X}_{i,t}$ includes the same controls as in Table 7.[23]

Table 9 reports the results. First, we note the results for mobile traders and day traders in columns 2, 3, 5, and 6. RTP predicts net buying near the open and net selling near the close, and

[21] In untabulated results, we find that the day-trade share of volume falls monotonically with account balance, from 21% in the smallest balance quintile to 14% in the largest, within every trading medium.

[22] Appendix C characterizes the compliers and formalizes the persistence mechanism. Consistent with the latter, the OLS–2SLS gap is smaller in untabulated results that use contemporaneous RTP as the regressor.

[23] In untabulated results, we confirm that coefficients are similar in a pooled panel regression with day and industry fixed effects and standard errors double-clustered by firm and day.

the pattern is as strong over the subsequent 5 and 20 days as on the next day. For mobile traders, a one-standard-deviation increase in RTP predicts 0.22bps higher net buying at the open and 0.10bps net selling at the close on the next day, and 0.22bps and 0.13bps per day over the next 20 days. These magnitudes appear small because the dependent variable is the net flow of a single hour scaled by shares outstanding. Day traders exhibit a similar pattern: 0.03bps higher net buying at the open and 0.02bps net selling at the close, both significant at the 1% level, again at every horizon. The persistence of opening net purchases and closing net sales over the following weeks suggests that RTP captures a recurring preference for daytime exposure among mobile traders and day traders. A short-lived episode of attention alone would not readily explain this pattern.

[Table 9 here]

Aggregate retail flow in columns 1 and 4 shows the buying at the open but not the selling at the close. Appendix Table A10 decomposes retail flow by trading medium. Mobile traders buy at the open and sell at the close, whereas investors trading through desktop and other channels are net buyers at the close in high-RTP stocks, so the closing sales of the mobile and day-trading clientele are absorbed in part within retail, by the more experienced and better-capitalized investors (Table 3). The aggregate therefore conceals the second leg of the round trip.

Price pressure from retail traders. Because net buy across investor types sums to zero within each stock and interval—for every buyer there is a seller—the net flow pattern alone does not immediately imply price pressure. We provide supplementary evidence that retail flows are "impatient." If retail investors demand immediacy, they should buy at higher prices and sell at lower prices relative to institutional traders who provide liquidity.

Figure 8 plots the average deviation of retail transaction prices from short-term institutional transaction prices within the same stock, day, and 30-minute interval, measured in basis points. We use short-term institutions as the reference because most market makers are classified into this category in the KRX data. The solid lines show the deviation of each group's buy prices from short-term institutional buy prices, and the dashed lines show the corresponding deviation of sell prices. The figure shows mobile retail traders and other retail traders.

[Figure 8 here]

Mobile traders exhibit the largest price impact. They buy at prices approximately 5.44bps higher than short-term institutions in the first half hour of trading and sell at prices approximately 4.50bps

lower in the final interval before the close. This pattern is consistent with mobile traders demanding immediacy and absorbing liquidity, thereby pushing prices up when they buy and down when they sell. Other retail traders exhibit a similar pattern that is weaker at the open (2.27bps) and comparable at the close (4.97bps). These findings align with two patterns presented earlier. First, mobile traders appear less sophisticated in the brokerage data, with smaller account balances, lower self-reported investment knowledge, and worse performance (Table 3). Second, mobile traders exhibit the strongest retail ebb and flow, with the most pronounced net buying at the open and net selling at the close (Figure 3). Together, these results suggest that mobile traders are a key source of intraday price pressure.

Appendix E, which examines the clienteles on the other side of these flows, shows the same story in aggressive order flow: relative to each group's all-day average, retail investors initiate more buys at the open and more sells at the close, most strongly among mobile traders (Figure A4). Short-term institutions and foreign investors take the other side, selling into the elevated open and buying toward the close. In Fama–MacBeth regressions, we interact RTP with an indicator for an above-median short-term institution trading proportion, foreign trading proportion, or share of retail's opening-half-hour buying executed against these two groups. Each interaction lowers the RTP coefficient by 0.62 to 0.79pp, against RTP coefficients of 2.81 to 2.97pp (Appendix Table A9).

4.2 Sensation-seeking and illusion of control

We next examine whether the preference for daytime exposure stems from behavioral patterns like sensation-seeking and illusion of control.

Sensation-seeking investors are attracted to volatile stocks and frequent trading (Grinblatt and Keloharju 2009). If day trading provides entertainment or excitement, investors with a taste for sensation may prefer to hold risky positions during market hours when they can monitor them, but close positions by day's end to avoid overnight exposure. Consistent with this hypothesis, the flows of this clientele chase past volatility. Appendix Table A11 reports the full coefficient estimates of the regressions in Table 9: stocks with high idiosyncratic volatility attract stronger net buying at the open and stronger net selling at the close among mobile traders and day traders, and the maximum daily return enters with the same signs. For aggregate retail flow, volatility predicts only the opening leg, for the reason given in Section 4.1.

Illusion of control—the belief that active monitoring and frequent trading improve outcomes—would reinforce this behavior. Investors who overestimate their ability to time intraday price

movements derive utility from the perception of active management (Barber and Odean 2001a). Mobile and desktop trading platforms amplify both mechanisms by providing real-time price updates and low-friction execution, making it easier to open and close positions within a single day.

For each investor, we compute the proportion of day-trade volume out of total trading volume (DT prop), with day trades defined as in Section 2.3. We then estimate the following cross-sectional regression:

$$\text{DT prop}_i = \alpha + \beta \mathbf{1}(\text{Male})_i + \gamma \mathbf{1}(\text{Young})_i + \delta \mathbf{1}(\text{High inv. know.})_i + \varepsilon_i, \tag{6}$$

where DT prop_i is investor i's proportion of day-trade positions (expressed in percentage points), $\mathbf{1}(\text{Young})_i$ equals one if investor i is aged 35 or below at the end of 2018, and $\mathbf{1}(\text{High inv. know.})_i$ equals one if investor i self-reported high investment knowledge.[24]

Table 10 presents the results. Column 3, which includes both the demographic and the trading-medium indicators, shows that male traders engage in 2.94pp more day trading than female traders, consistent with Barber and Odean (2001a), who document that men trade more frequently due to overconfidence. Younger traders (aged 35 or below) engage in 3.68pp more day trading, and those with high self-reported investment knowledge engage in 5.19pp more. MTS users engage in 6.06pp more day trading than users of other media, and HTS users engage in 13.05pp more. Columns 1 and 2 show that each set of effects is similar without the other set of controls. Column 4 includes interactions. The day-trading propensity is highest among young male traders with high self-reported knowledge, with the three-way interaction adding 2.21pp. Overall, we see that day trading is concentrated among young, male, technology-enabled traders who self-report high investment knowledge.

[Table 10 here]

4.3 Alternative and complementary mechanisms

We do not claim that daytime exposure is the only driver of the return gap. We view many alternative explanations as potentially reinforcing rather than competing with our mechanism. We assess this in two ways. First, we test whether RTP's predictive power survives controls for variables proposed in prior studies. Second, we examine whether RTP's effect varies across subsamples where alternative

[24]The available responses ranged from 1 (low) to 4 (high) on an ordinal scale. We define "high" as a response of 4 and code all other responses as zero. See Appendix Table A1 for details.

mechanisms should be stronger or weaker.

Robustness tests. We begin with Fama–MacBeth regressions that progressively add controls motivated by alternative explanations. We include all controls in Table 7 as well.

Table 11 presents the results. Column 1 reproduces our baseline specification from Table 7, which already includes idiosyncratic volatility among the controls. Idiosyncratic volatility captures retail attention as in Berkman et al. (2012) and reflects limits to arbitrage as documented in Stambaugh et al. (2015).

[Table 11 here]

Column 2 adds abnormal volume from Barber and Odean (2008), computed as the difference between turnover in month t and average turnover over the prior 12 months. This variable provides an additional proxy for attention-driven trading activity beyond what idiosyncratic volatility captures.

Column 3 adds overnight systematic variance from Bogousslavsky (2021). Following his approach, we estimate firm-specific overnight betas by regressing each stock's overnight return on the market overnight return using a rolling 12-month window with at least 100 daily observations. Overnight systematic variance is then computed as the product of squared beta and variance of market returns estimated from the regressions, capturing exposure to systematic overnight risk.

Column 4 adds retail order imbalance at the open from Lu et al. (2023). We compute this at the daily level as the buy-minus-sell volume of aggressive retail orders during the first hour of trading (9:00–10:00, excluding the opening auction), scaled by twice the total daily volume, then aggregate to the monthly level. In Lu et al. (2023), high retail opening order imbalance allows fast arbitrageurs to absorb inventory during periods of high information asymmetry when cream-skimming risk prevents competition from lower-cost slow arbitrageurs. Consequently, fast arbitrageurs become the marginal liquidity providers and set opening prices that deviate from fundamental values to compensate for their high inventory costs. Thus, high retail buying pressure at the open predicts positive overnight returns and the associated night-minus-day wedge.

Column 5 includes overnight beta from Hendershott et al. (2020), estimated using the same rolling regression as overnight systematic variance but requiring only 30 daily observations. This beta captures differential sensitivity to overnight versus intraday market movements.

Column 6 adds abnormal night reversal from Akbas et al. (2022). For each stock-month, we compute the proportion of trading days with a positive overnight return followed by a negative

intraday return, and abnormal night reversal is this proportion divided by its average over the trailing 12 months, capturing abnormal intensification of the tug-of-war pattern.

Across all specifications, RTP remains statistically significant and economically large. In the fully saturated specification in column 6, the coefficient on RTP is 0.94 for overnight returns and −1.07 for intraday returns, close to 1.05 and −1.26 in the baseline specification (Table 7). This indicates that RTP captures variation in the return gap beyond what these alternative mechanisms explain, though the modest attenuation suggests some overlap with complementary channels.

Several alternative mechanisms exhibit significant coefficients, consistent with their proposed roles in generating the overnight–intraday return gap. Column 3 shows that overnight systematic risk increases the return gap, as expected from Bogousslavsky (2021). Column 4 shows that the fast- and slow-arbitrageur mechanism from Lu et al. (2023) also plays a role in increasing the return gap. Some variables are absorbed by others as they capture similar concepts. For instance, the effect of abnormal volume in column 2 becomes small once other attention proxies are included, while overnight beta in column 5 enters with expected signs but modest magnitudes.

Heterogeneous effects. Limits to arbitrage, for example, make it costlier for arbitrageurs to trade against predictable price pressure: idiosyncratic volatility raises the risk of holding a position against the crowd (Pontiff 2006, Stambaugh et al. 2015), and overnight risk raises the cost of carrying it through the night (Bogousslavsky 2021). If these complementarities hold, we should observe stronger RTP effects in subsamples where these mechanisms are most active.

Table 12 presents results from Fama–MacBeth regressions that interact RTP with indicator variables for high values of each mechanism variable:

$$y_{i,t+1} = \alpha_t + \beta_t \mathrm{RTP}_{i,t} + \gamma_t \mathrm{High}_{i,t} + \delta_t \mathrm{RTP}_{i,t} \times \mathrm{High}_{i,t} + \mathbf{X}'_{i,t}\Gamma_t + \varepsilon_{i,t+\tau} \tag{7}$$

where $\mathrm{High}_{i,t} = \mathbf{1}[Z_{i,t} > \mathrm{median}_t(Z_{i,t})]$ indicates whether the mechanism variable $Z_{i,t}$ exceeds its cross-sectional median in month t. The parameter of interest is δ_t, which captures whether RTP's effect on the return gap is amplified (if positive) or attenuated (if negative) when the alternative mechanism is strong. Columns 1–6 enter one interaction at a time, and column 7 enters all six jointly.

[Table 12 here]

The interaction terms show where RTP's effect is stronger. Each variable does not exhaustively or

exclusively capture a single concept. For instance, idiosyncratic volatility proxies for both attention and limits to arbitrage, and both overnight systematic variance and overnight beta capture overnight risk. Thus, we cannot interpret every single coefficient in isolation. Two interactions nonetheless stand out. The interaction with idiosyncratic volatility in column 1 is positive for overnight returns and negative for intraday returns, both significant at the 1% level, and the interaction with retail order imbalance at the open in column 4 has the same signs and significance. Both survive when all six interactions enter jointly in column 7.

For the return gap, the ratio of each interaction coefficient to the RTP coefficient in the same column measures the amplification. RTP's effect is 37% larger among high-volatility stocks (0.79 relative to 2.14 in column 1 of Appendix Table A13) and 22% larger where retail opening order imbalance is high (0.47 relative to 2.17 in column 4). The remaining interactions are small relative to the RTP coefficient and at most marginally significant. The high indicators for overnight systematic variance and overnight beta nearly coincide, since both derive from the same rolling regression, so their interactions are not separately identified in column 7. We read this as evidence that the retail ebb and flow is complementary with attention, limits to arbitrage, and the liquidity-provision channel of Lu et al. (2023). We find no comparable amplification by systematic overnight risk, although stock-specific overnight risk may still limit arbitrage.

5 Conclusion

Our results show that retail trading plays a central role in generating the overnight–intraday return gap. Leveraging trade-level data from the KRX that explicitly identifies investor types, we document a robust positive relationship between RTP and the return gap. Using nominal share price as an instrument for retail participation, we provide causal evidence that higher retail participation widens the gap.

We attribute this phenomenon to a recurrent trading pattern we term retail ebb and flow. Retail investors, particularly those engaging in day trading via mobile platforms, systematically exhibit net buying pressure at the market open and net selling pressure near the close. This behavior is consistent with a preference for daytime exposure driven by sensation-seeking and the illusion of control. Investors seek the utility of monitoring and managing positions during active market hours but prefer to close positions before the overnight period when they cannot act. Our analysis of account-level brokerage data confirms that this intraday turnover is concentrated among young,

male investors and those using technology-enabled trading interfaces.

Our findings demonstrate that retail trading exerts an economically significant and persistent influence on the intraday pattern of stock returns. This insight is particularly timely given the global surge in retail participation and the proliferation of mobile trading platforms.

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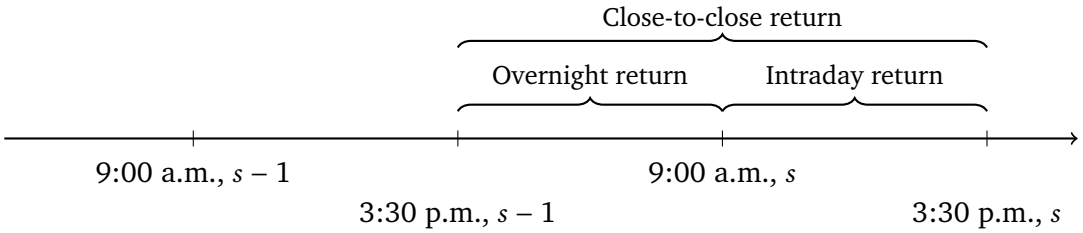


Figure 1. Overnight and intraday returns

This figure illustrates the decomposition of daily stock returns. We define the overnight return as the close-to-open return from 3:30 p.m. on day $s-1$ to 9:00 a.m. on day s, and the intraday return as the open-to-close return from 9:00 a.m. to 3:30 p.m. on day s. These two components combine to form the close-to-close return. The Korean stock market opens at 9:00 a.m. and closes at 3:30 p.m. Before August 1, 2016, the market closed at 3:00 p.m.

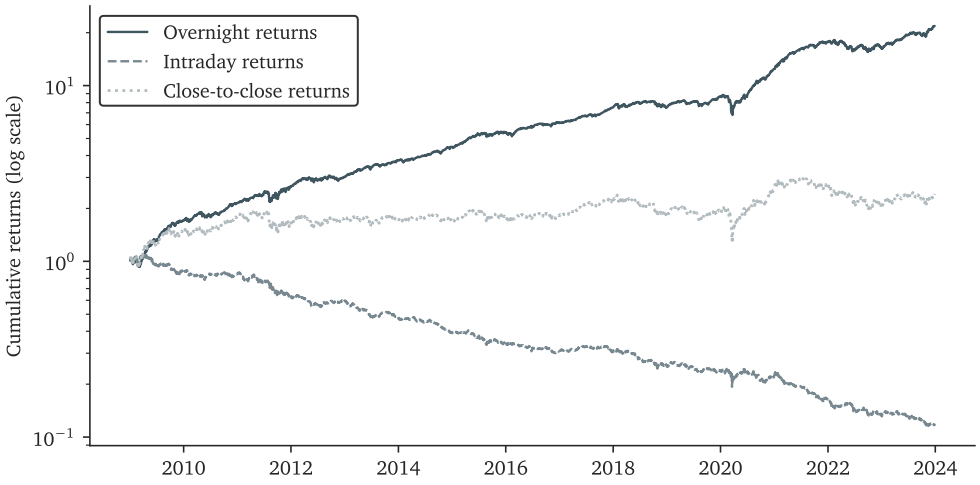
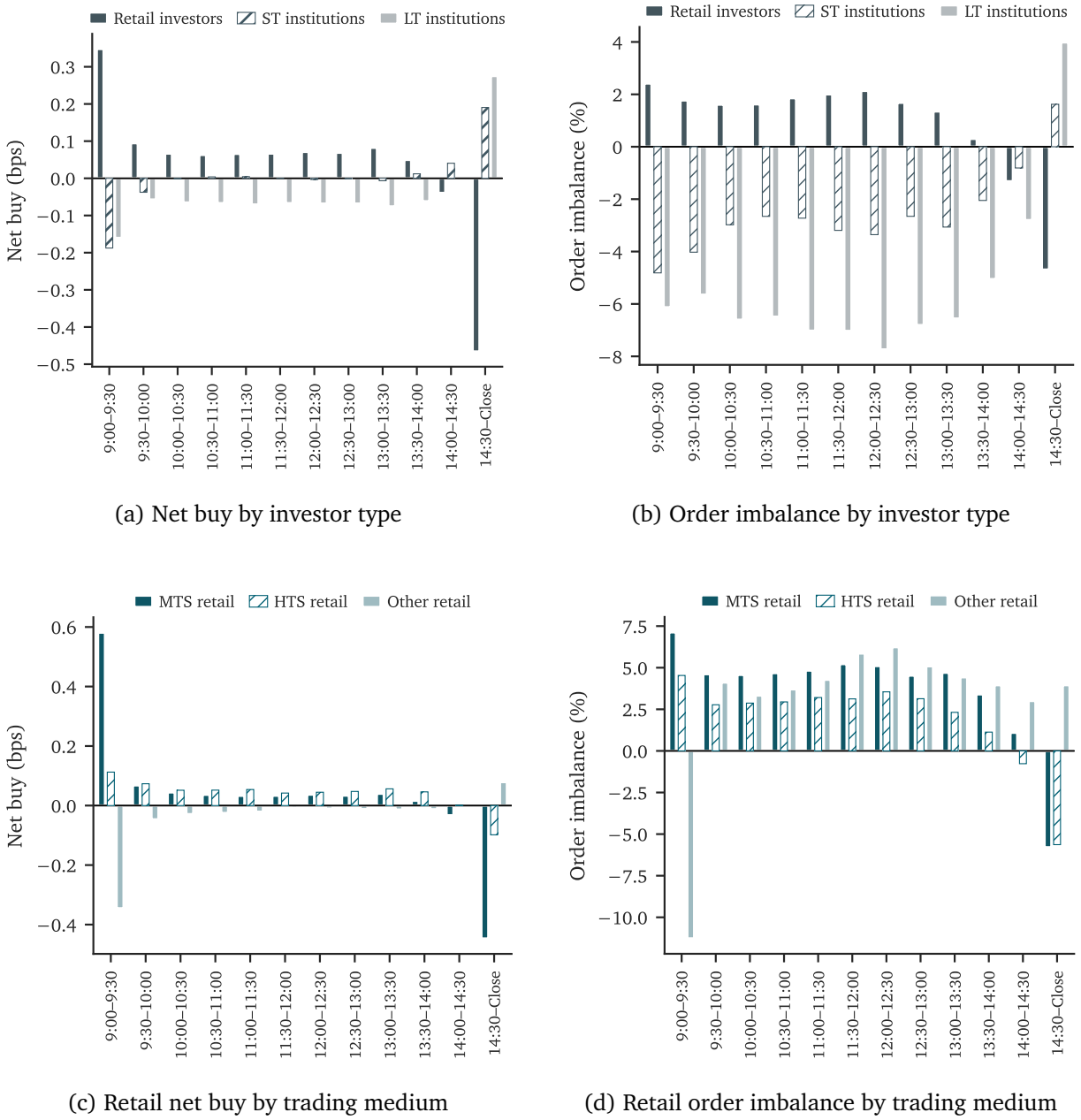


Figure 2. Cumulative overnight and intraday returns

This figure plots cumulative value-weighted returns for all common stocks listed on the Korean stock market from January 2009 to December 2023 on a log scale. We compute daily overnight returns (close-to-open), intraday returns (open-to-close), and close-to-close returns for each stock, then aggregate using the previous day's market capitalization as weights. A $1 investment in the overnight component grows to approximately $21.84, while the same investment in the intraday component declines to $0.12.



(a) Net buy by investor type

(b) Order imbalance by investor type

(c) Retail net buy by trading medium

(d) Retail order imbalance by trading medium

Figure 3. Intraday net buy and order imbalance patterns

This figure plots average retail and institutional trading flows across all stocks and trading days for each 30-minute interval of the session. Panels (a) and (c) measure flows as net buy: buy volume minus sell volume, in basis points of shares outstanding. Panels (b) and (d) measure the same flows as order imbalance: buy volume minus sell volume divided by buy plus sell volume within the stock, day, and interval, in percent, computed for stock-day-intervals in which the group trades. Panels (a) and (b) show retail investors, short-term (ST) institutions, and long-term (LT) institutions. Panels (c) and (d) decompose retail flows by trading medium: mobile trading system (MTS), home trading system (HTS), and other media. The first interval covers 9:00 to 9:30 and includes the opening auction. The last interval runs from 14:30 to the market close (15:00 before August 2016 and 15:30 afterwards) and includes the closing auction. The sample period is January 2009 to December 2023.

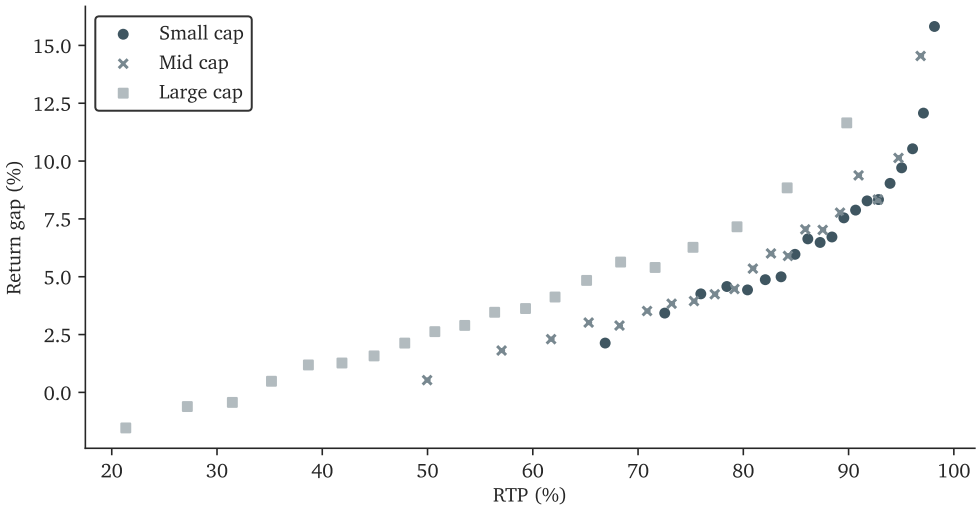


Figure 4. RTP and the overnight–intraday return gap

This figure plots binned scatter plots of the monthly return gap against retail trading proportion (RTP) within size terciles. Each month, we assign stocks to terciles based on average daily market capitalization, then create 20 bins of RTP within each tercile. Each point represents the average return gap and average RTP within a bin. The return gap is the monthly gap RG^m defined in Section 2.2, expressed in percentage points. RTP is the ratio of retail trading volume to total trading volume within the month, expressed in percent. The sample period is January 2009 to December 2023.

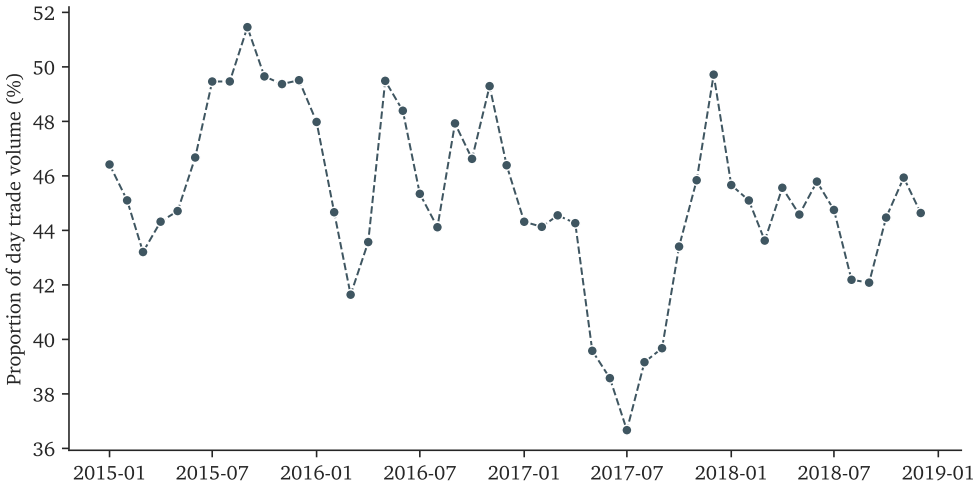
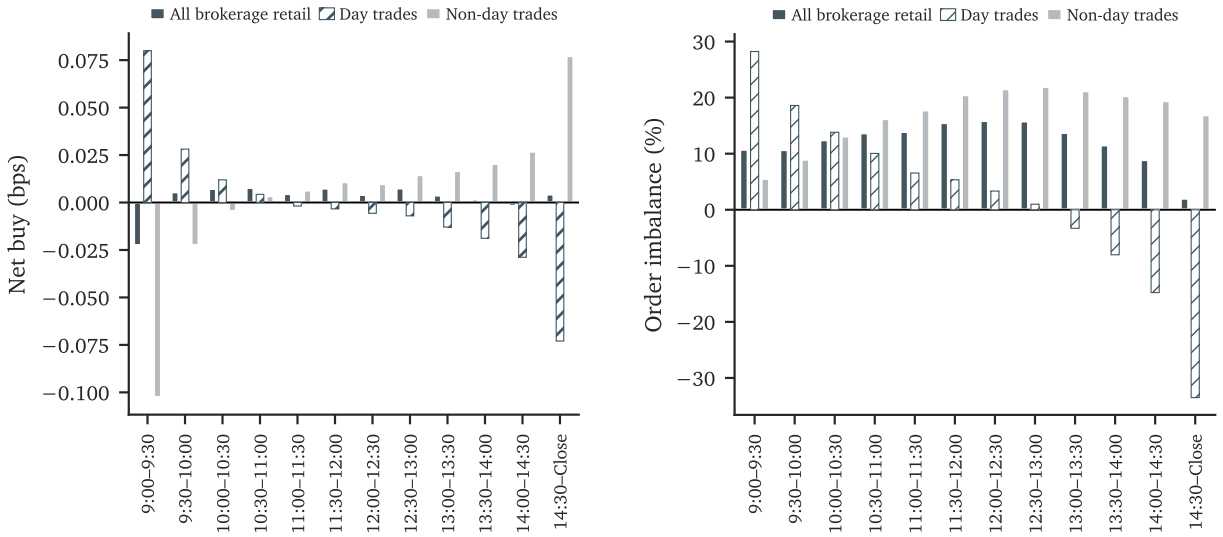


Figure 5. Proportion of day trade volume in the brokerage dataset

This figure plots the monthly proportion of trading volume attributable to day trades in the brokerage dataset. For each investor, stock, and day, we classify the investor's volume in that stock as day-trade volume if the investor both buys and sells the stock within the same trading session. The proportion is computed as the total value of day trade volume divided by total trading volume in each month, expressed in percent. The sample covers approximately 235,000 active retail accounts at a large Korean discount brokerage from January 2015 to December 2018.

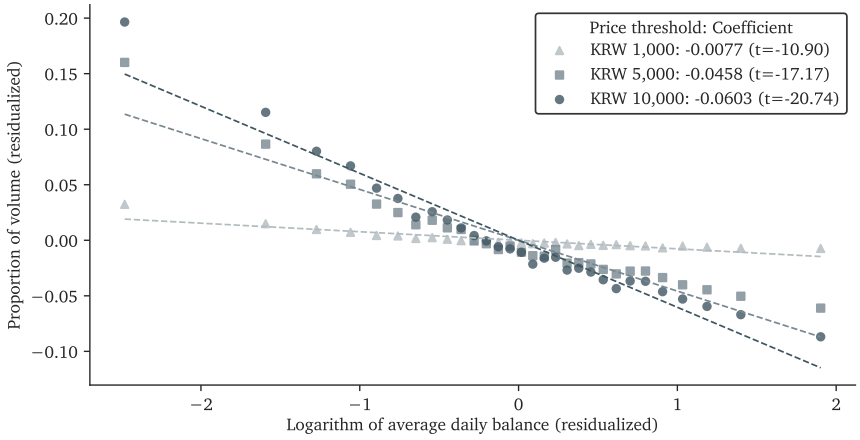


(a) Net buy

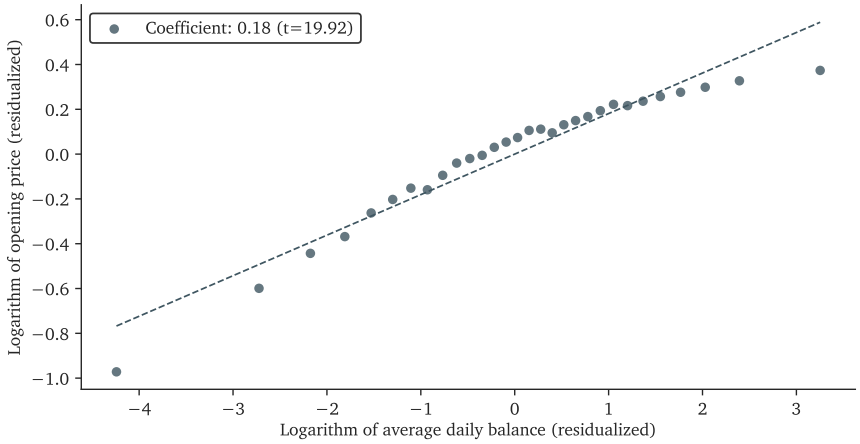
(b) Order imbalance

Figure 6. Intraday net buy and order imbalance: Day trades vs. non-day trades

This figure plots average brokerage retail flows across stocks and trading days for each 30-minute interval, in total and split into day-trade and non-day-trade components, constructed in parallel with Figure 3. Day trades are an investor's trades in a stock on a day on which the investor both buys and sells that stock. Non-day trades are all other trades, including positions carried overnight and closed on a later day. Panel (a) measures flows as net buy in basis points of shares outstanding, which scales with the group's share of trading volume. Panel (b) measures the same flows as order imbalance in percent of the group's own buy plus sell volume, which is invariant to the group's size. The sample covers the large-cap universe of Figure 3, and intervals follow that figure, with the first including the opening auction and the last running from 14:30 to the market close. The sample covers the brokerage dataset from January 2015 to December 2018.



(a) Proportion of volume in low-price stocks



(b) Volume-weighted average opening price

Figure 7. Tendency to trade low-price stocks and daily balance

This figure plots binned scatter plots of the relationship between average daily account balance and the tendency to trade low-price stocks, controlling for investor demographics. For each investor in the brokerage dataset, we compute the average daily account balance over the sample period and two measures of low-price stock trading: (a) the proportion of trading volume (in KRW) executed in stocks with opening prices below KRW 1,000, 5,000, and 10,000, and (b) the logarithm of the volume-weighted average opening price across all stocks traded. We residualize both variables with respect to sex, age, income, investment knowledge, derivative experience, and trading medium, then create 30 bins based on residualized log average daily balance. Each point represents the average residualized values within a bin. The dashed lines show fitted linear relationships, with coefficients and t-statistics reported in the legends. The sample covers the approximately 148,000 active retail accounts with a positive average balance and complete questionnaire responses, from January 2015 to December 2018.

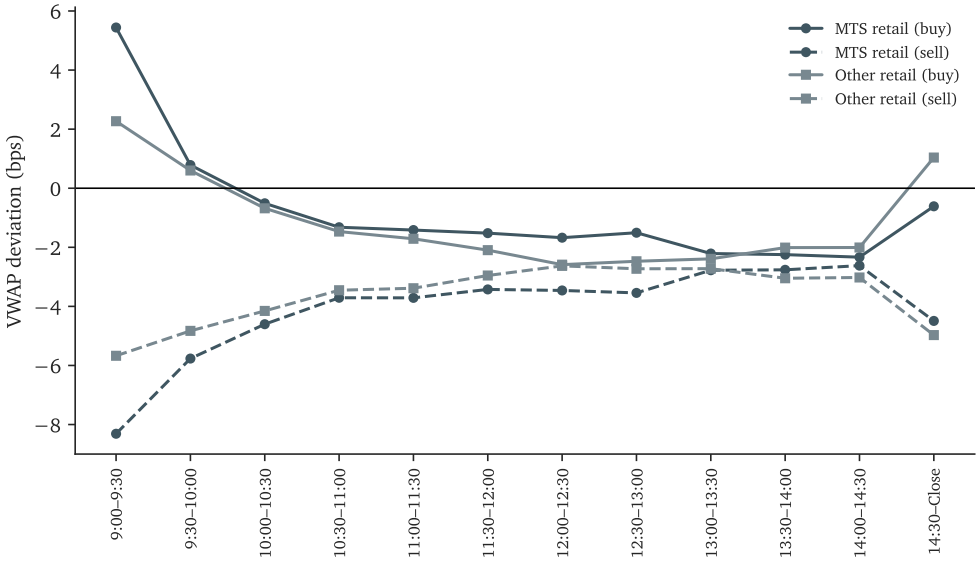


Figure 8. Intraday VWAP deviation relative to short-term institutions

This figure plots the average deviation of retail transaction prices from short-term institutional transaction prices within the same stock, day, and 30-minute interval, measured in basis points. For each 30-minute interval, we compute the volume-weighted average price (VWAP) separately for mobile retail traders (MTS), other retail traders, and short-term institutions. The solid lines show the deviation of each group's buy prices from short-term institutional buy prices, and the dashed lines show the corresponding deviation of sell prices. The first interval, 9:00–9:30, includes the opening auction, and the last interval runs from 14:30 to the market close (15:00 before August 2016 and 15:30 afterwards) and includes the closing auction. The sample period is January 2009 to December 2023.

Table 1. Summary statistics for the KRX data

This table presents summary statistics for the main variables used in the analysis. The sample consists of 169,961 stock-month observations covering common stocks listed on the Korea Exchange from January 2009 to December 2023, restricted to the largest half of stocks by beginning-of-year market capitalization. Overnight return is the close-to-open return, intraday return is the open-to-close return, both compounded within the month, and return gap is the monthly gap RG^m defined in Section 2.2. Retail trading proportion (RTP) is the ratio of retail trading volume to total trading volume over all trading days in the month. Nominal price is the share price on the first trading day of the month, the instrument for RTP in Table 8, in thousands of KRW. Other variables include standard firm characteristics (size, book-to-market, beta, illiquidity, foreign ownership) and past return measures (120-day momentum, idiosyncratic volatility, maximum daily return). Detailed variable definitions are provided in Appendix Table A1. Summary statistics for the variables specific to other specifications are reported in Appendix Table A14.

	N	Mean	SD	Min	25th	50th	75th	Max
Overnight return (%)	169,961	2.37	8.41	-76.05	-2.26	1.68	6.06	205.36
Intraday return (%)	169,961	-1.45	11.96	-62.49	-8.44	-2.12	4.51	238.95
Return gap (%)	169,961	5.52	16.88	-76.63	-4.68	3.80	13.46	370.84
RTP (%)	169,961	72.98	20.33	4.00	60.79	78.54	88.99	100.00
Nominal price (thousand)	169,961	41.82	113.45	0.08	6.12	13.55	34.40	3337.00
Log market equity	169,961	5.89	1.32	1.92	4.98	5.57	6.50	13.11
Book-to-market ratio	169,961	0.99	0.89	0.00	0.39	0.74	1.32	15.19
Market beta	169,961	0.89	0.50	-3.56	0.54	0.87	1.22	5.25
Amihud illiquidity	169,961	0.07	0.91	0.00	0.00	0.01	0.03	137.43
Foreign ownership (%)	169,961	10.31	13.36	0.00	1.59	5.10	13.67	99.25
Past 120-day return (%)	169,961	4.05	38.20	-92.72	-17.09	-2.72	15.73	1244.74
Idiosyncratic volatility (%)	169,961	2.10	1.22	0.18	1.32	1.82	2.51	18.79
Maximum daily return (%)	169,961	5.96	4.47	0.00	3.16	4.75	7.17	30.00

Table 2. Cross-sectional correlation among firm characteristics

This table presents time-series averages of cross-sectional correlations among the main variables. For each month from January 2009 to December 2023, we compute pairwise correlations across all stocks in the sample, then average these correlations over time. The sample consists of common stocks listed on the Korea Exchange, restricted to the largest half of stocks by beginning-of-year market capitalization. Variable definitions are provided in Appendix Table A1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) Overnight return (%)											
(2) Intraday return (%)	-0.22										
(3) Return gap (%)	0.67	-0.84									
(4) RTP (%)	0.20	-0.10	0.19								
(5) Log market equity	-0.03	0.10	-0.10	-0.74							
(6) Book-to-market ratio	-0.09	0.07	-0.11	-0.06	-0.10						
(7) Market beta	0.08	-0.08	0.10	0.12	0.07	-0.14					
(8) Amihud illiquidity	-0.02	0.03	-0.02	0.11	-0.19	0.18	-0.19				
(9) Foreign ownership (%)	-0.07	0.04	-0.08	-0.58	0.54	0.02	-0.05	-0.07			
(10) Past 120-day return (%)	0.20	0.27	-0.08	-0.02	0.15	-0.14	-0.07	-0.03	0.02		
(11) Idiosyncratic volatility (%)	0.31	0.13	0.11	0.31	-0.12	-0.22	0.15	-0.08	-0.16	0.25	
(12) Maximum daily return (%)	0.33	0.27	0.00	0.29	-0.08	-0.18	0.18	-0.07	-0.14	0.25	0.86

Table 3. Summary statistics for the brokerage dataset

This table presents summary statistics for approximately 235,000 active retail accounts at a large Korean discount brokerage from January 2015 to December 2018. Active accounts are those that executed at least 10 trades over the sample period. The brokerage provided data on all such accounts. We exclude accounts without a computable realized return and account holders younger than 20 or older than 85. Panel A reports statistics for all investors. Panel B reports statistics by main trading medium: mobile (MTS), desktop (HTS), and other traditional channels (Other) such as brokerage-assisted or web-based trading. Main trading medium is the interface through which the investor executes the most trades. Performance is realized return over the sample period, winsorized at 1% and 99%. Day trade volume proportion is the percentage of trading volume attributable to stock-days on which the investor both buys and sells the stock. Average daily balance and average trade size are reported in millions of KRW, and performance and the day trade volume proportion are in percent. Approximately half of investors completed questionnaires on monthly income, investment knowledge, investment aggressiveness, and derivative experience. Detailed variable definitions are provided in Appendix Table A1.

Panel A: All investors

	N	Mean	SD	Min	25th	50th	75th	Max
Male	235,445	0.61	0.49	0.00	0.00	1.00	1.00	1.00
Age	235,445	46.15	11.57	20.00	37.00	44.00	54.00	85.00
Monthly income	143,119	2.86	1.06	0.00	2.00	3.00	3.00	5.00
Investment knowledge	143,119	2.53	0.93	0.00	2.00	2.00	3.00	4.00
Investment aggressiveness	143,119	3.99	0.94	1.00	4.00	4.00	5.00	5.00
Derivative experience	143,119	1.68	0.86	0.00	1.00	1.00	3.00	3.00
Performance	235,445	-4.07	20.03	-70.44	-9.51	-1.95	1.26	98.07
Avg. daily balance	235,445	42.10	291.42	-38.85	3.89	11.48	32.08	50,364
Avg. trade size	235,445	4.24	8.40	0.00	0.73	2.00	4.84	778.27
Number of unique stocks traded	235,445	47.31	94.03	1.00	8.00	18.00	46.00	2,215
Number of trades	235,445	796.31	7,578	10.00	32.00	98.00	352.00	1,379,813
Day trade volume proportion	235,445	17.01	21.91	0.00	0.00	7.57	26.58	100.00

Panel B: Investors by trading medium

	MTS		HTS		Other	
	N	Mean	*N*	Mean	*N*	Mean
Male	125,498	0.64	52,485	0.65	57,455	0.48
Age	125,498	42.00	52,485	49.84	57,455	51.83
Monthly income	67,658	2.78	30,332	3.01	45,122	2.89
Investment knowledge	67,658	2.36	30,332	2.76	45,122	2.64
Investment aggressiveness	67,658	3.74	30,332	4.16	45,122	4.24
Derivative experience	67,658	1.54	30,332	1.81	45,122	1.79
Performance	125,498	-4.75	52,485	-3.13	57,455	-3.44
Avg. daily balance	125,498	23.26	52,485	68.68	57,455	59.00
Avg. trade size	125,498	3.20	52,485	4.60	57,455	6.18
Number of unique stocks traded	125,498	40.32	52,485	78.10	57,455	34.46
Number of trades	125,498	473.99	52,485	2,074	57,455	333.36
Day trade volume proportion	125,498	17.11	52,485	24.11	57,455	10.31

Table 4. Time-series evidence

This table presents the time-series relation between aggregate retail trading proportion (RTP) and subsequent market returns at daily and monthly frequencies. Panel A reports average returns following high-RTP and low-RTP periods, where high-RTP periods are those with aggregate RTP above the sample median. Panel B reports regression coefficients from predictive regressions of next-period market returns on current-period aggregate RTP, controlling for contemporaneous market returns and market volatility. Stock-days with an overnight, intraday, or close-to-close return outside ±50% are excluded from the market aggregates. Returns are in percentage points and RTP is a fraction, so the coefficients indicate the change in returns (in basis points) associated with a one-percentage-point increase in RTP. For daily (monthly) observations, Newey–West standard errors are reported in parentheses, computed with 20 lags (six lags). The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Daily			Monthly		
	Overnight	Intraday	Return gap	Overnight	Intraday	Return gap
Panel A: Average returns following periods of high and low RTP						
High RTP	0.11***	-0.06***	0.18***	2.40***	-1.56***	4.10***
	(0.02)	(0.02)	(0.02)	(0.53)	(0.32)	(0.58)
Low RTP	0.06***	-0.04**	0.10***	1.20***	-0.76***	2.05***
	(0.01)	(0.02)	(0.02)	(0.28)	(0.26)	(0.34)
High – Low	0.05**	-0.02	0.08**	1.19*	-0.80*	2.06***
	(0.02)	(0.03)	(0.03)	(0.64)	(0.47)	(0.69)
Panel B: Predictive regressions with RTP						
RTP	0.39**	-0.31	0.70***	9.05**	-8.41**	18.07***
	(0.18)	(0.19)	(0.23)	(4.06)	(4.14)	(4.12)
Controls	✓	✓	✓	✓	✓	✓
N	3,686	3,686	3,686	178	178	178
Adjusted R^2 (%)	0.60	0.07	0.14	13.09	5.60	12.84

Table 5. Portfolios sorted by retail trading proportion (RTP)

This table presents the monthly average returns of portfolios sorted by retail trading proportion (RTP). Each month, stocks are sorted into deciles based on RTP. Panels A and B report overnight and intraday returns in excess of the risk-free rate, and Panel C reports the return gap, for each decile portfolio. H–L is a long-short portfolio that buys the highest RTP decile and sells the lowest RTP decile. α_{FF5} denotes alphas from the Fama–French five-factor model estimated for the Korean market. Newey–West t-statistics are reported in brackets, computed with six lags. Portfolios are value-weighted and rebalanced monthly. The sample period is January 2009 to December 2023.

	Low	2	3	4	5	6	7	8	9	High	H-L
Panel A: Overnight return											
Mean	0.89	1.89	2.67	2.93	3.41	3.49	3.45	3.72	3.96	5.27	4.37
	[2.22]	[4.04]	[5.43]	[5.42]	[6.43]	[6.07]	[6.03]	[6.58]	[6.45]	[7.90]	[8.79]
α_{FF5}	0.70	1.66	2.38	2.59	3.11	3.13	3.07	3.44	3.54	4.94	4.24
	[2.93]	[6.73]	[8.76]	[10.12]	[11.99]	[10.50]	[9.70]	[11.28]	[10.20]	[11.57]	[10.20]
Panel B: Intraday return											
Mean	-0.30	-1.48	-1.76	-2.62	-2.34	-2.81	-2.74	-3.10	-3.32	-5.47	-5.16
	[-0.91]	[-4.54]	[-5.29]	[-7.64]	[-5.96]	[-7.94]	[-7.21]	[-8.62]	[-10.26]	[-12.02]	[-9.58]
α_{FF5}	-0.64	-1.81	-2.02	-2.91	-2.54	-3.04	-2.85	-3.22	-3.47	-5.57	-4.92
	[-3.13]	[-7.21]	[-7.70]	[-8.32]	[-7.19]	[-10.11]	[-7.10]	[-9.04]	[-10.06]	[-12.69]	[-10.22]
Panel C: Overnight-intraday return gap											
Mean	1.89	4.36	5.67	7.05	7.57	8.15	8.29	9.04	9.65	14.53	12.64
	[4.88]	[9.23]	[11.47]	[10.25]	[11.46]	[12.41]	[11.67]	[12.91]	[12.04]	[15.15]	[13.10]
α_{FF5}	2.04	4.48	5.67	7.00	7.51	8.02	8.01	8.90	9.37	14.34	12.29
	[4.85]	[9.40]	[10.65]	[10.83]	[11.97]	[12.93]	[11.92]	[12.61]	[12.98]	[15.38]	[13.19]

Table 6. Return gaps of RTP-sorted portfolios controlling for size and volatility

This table presents average monthly return gaps of portfolios sorted by retail trading proportion (RTP) conditional on market equity (Panel A) or past return volatility (Panel B). Each month, stocks are first sorted into quintiles based on the control variable, then sorted into quintiles based on RTP within each control quintile. H–L is a long-short portfolio that buys the highest RTP quintile and sells the lowest RTP quintile within each control group. Newey–West t-statistics are reported in brackets for H–L portfolios. Portfolios are value-weighted and rebalanced monthly. The sample period is January 2009 to December 2023.

	Low RTP	2	3	4	High RTP	H-L	t-statistic
Panel A: Double sort on RTP and market cap							
Small	2.56	4.92	5.79	7.17	9.24	6.68	[9.32]
P2	2.47	4.83	5.83	7.78	12.36	9.89	[11.65]
P3	1.18	4.41	6.03	7.85	12.87	11.70	[11.82]
P4	-0.28	2.75	5.13	7.74	11.52	11.80	[16.52]
Big	1.51	2.82	4.18	6.46	9.73	8.23	[11.55]
Panel B: Double sort on RTP and return volatility							
Low Vol	0.77	2.61	2.22	3.02	3.59	2.82	[4.77]
P2	2.10	4.11	4.17	5.21	6.23	4.13	[7.25]
P3	2.71	5.64	6.16	6.77	8.14	5.43	[6.89]
P4	4.58	8.04	9.32	8.99	10.78	6.20	[8.94]
High Vol	7.77	11.11	13.05	13.99	19.31	11.54	[10.75]

Table 7. Fama–MacBeth regressions

This table presents results from Fama–MacBeth regressions of one-month-ahead returns on retail trading proportion (RTP). The dependent variables are one-month-ahead overnight returns in columns 1–3, one-month-ahead intraday returns in columns 4–6, and one-month-ahead return gaps in columns 7–9. Returns are expressed in percentage points. The independent variable of interest is RTP. Detailed variable definitions are provided in Appendix Table A1. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Newey–West standard errors are reported in parentheses, computed with six lags. Industry fixed effects include 51 KRX industry classifications. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Overnight return			Intraday return			Return gap		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
RTP	1.24***	1.05***	1.00***	-1.22***	-1.26***	-1.22***	2.97***	2.43***	2.35***
	(0.13)	(0.10)	(0.09)	(0.11)	(0.11)	(0.11)	(0.23)	(0.18)	(0.18)
Overnight return		1.16***	1.14***		-1.15***	-1.21***		2.51***	2.55***
		(0.08)	(0.08)		(0.08)	(0.08)		(0.17)	(0.18)
Intraday return		-0.19***	-0.24***		0.17**	0.07		-0.26**	-0.20*
		(0.05)	(0.04)		(0.08)	(0.08)		(0.12)	(0.11)
Log market equity		0.33***	0.42***		-0.71***	-0.76***		0.99***	1.15***
		(0.05)	(0.05)		(0.11)	(0.11)		(0.11)	(0.13)
Market beta		0.34***	0.30***		-0.23***	-0.22***		0.60***	0.54***
		(0.06)	(0.07)		(0.08)	(0.08)		(0.13)	(0.13)
Book-to-market		-0.25***	-0.16***		0.25***	0.30***		-0.64***	-0.60***
		(0.04)	(0.04)		(0.08)	(0.07)		(0.08)	(0.09)
Amihud illiquidity		0.04	0.03		0.17***	0.14**		0.07	0.08
		(0.04)	(0.04)		(0.06)	(0.06)		(0.10)	(0.11)
Foreign ownership		0.11***	0.08***		-0.08**	-0.11***		0.14***	0.15***
		(0.03)	(0.03)		(0.03)	(0.03)		(0.05)	(0.05)
Past 120-day return		0.21***	0.17***		-0.02	-0.03		0.39***	0.38***
		(0.04)	(0.03)		(0.09)	(0.08)		(0.11)	(0.10)
Idiosyncratic volatility		0.43***	0.47***		-1.09***	-1.05***		2.31***	2.29***
		(0.07)	(0.07)		(0.13)	(0.12)		(0.19)	(0.20)
Maximum daily return		0.04	0.02		0.29***	0.32***		-0.34**	-0.41**
		(0.05)	(0.06)		(0.09)	(0.10)		(0.14)	(0.16)
Industry FE			✓			✓			✓
N	169,961	169,961	159,472	169,961	169,961	159,472	169,961	169,961	159,472
Adjusted R^2 (%)	4.37	12.63	15.69	2.06	9.38	12.81	4.11	13.74	16.02

Table 8. 2SLS regressions with nominal price as instrument

This table presents results from panel OLS and two-stage least squares (2SLS) regressions of one-month-ahead returns on retail trading proportion (RTP). Column 1 reports first-stage results using the log of the nominal closing price on the first trading day of month t as an instrument for RTP. KP F-stat is the Kleibergen–Paap rank Wald F-statistic testing for weak identification. Columns 2–7 report second-stage and OLS results with one-month-ahead overnight returns, intraday returns, and return gaps as dependent variables, expressed in percentage points. Control variables are the same as those in Table 7. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Standard errors are reported in parentheses, double-clustered by firm and month. Industry fixed effects include 51 KRX industry classifications, and stock-months without an industry classification are excluded. R^2 within is computed after absorbing the fixed effects. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	RTP	Overnight return		Intraday return		Return gap	
	First stage	OLS	2SLS	OLS	2SLS	OLS	2SLS
RTP		0.99***	2.28***	-1.24***	-1.90***	2.36***	4.51***
		(0.09)	(0.24)	(0.11)	(0.30)	(0.16)	(0.45)
Nominal price	-0.19***						
	(0.01)						
Controls	✓	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓	✓
N	159,472	159,472	159,472	159,472	159,472	159,472	159,472
KP F-statistic	318.43						
R^2 within (%)	65.33	5.96	4.98	3.25	3.15	8.15	7.58

Table 9. Fama–MacBeth regressions: RTP predicting net buy flows

This table presents results from Fama–MacBeth regressions of future net buy flows on retail trading proportion (RTP) using daily stock-level observations. The dependent variables are investor net buy flows (buy minus sell volume, scaled by shares outstanding) measured in basis points. Columns 1–3 measure net buy using trades executed in the first hour of trading and columns 4–6 use trades executed in the last hour of trading. Column headers indicate the investor group: "Retail trader" includes all retail investors, "Mobile trader" includes retail investors using mobile trading systems (MTS), and "Day trader" includes round-trip positions opened and closed (completely or partially) within the same day. Retail trader and Mobile trader flows are based on KRX data, and Day trader flows are based on brokerage data. Panels A, B, and C report results when the dependent variable is the average net buy flow over the next 1, 5, and 20 days, respectively, following day t. RTP is measured on day t. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation, and the dependent flows are winsorized at the 1% and 99% levels within day. Coefficients are scaled by 100 for readability. A coefficient of 8.10 indicates that a one-standard-deviation increase in RTP is associated with a 0.0810 basis point increase in net buy flows. Newey–West standard errors are reported in parentheses, computed with 20 lags. The sample period is January 2009 to December 2023 for retail and mobile trades, and January 2015 to December 2018 for day trades. Appendix Table A11 reports the full coefficient estimates for Panel A. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Net buy at open			Net buy at close		
	(1) Retail	(2) Mobile	(3) Day trader	(4) Retail	(5) Mobile	(6) Day trader
Panel A: Flows on the next day						
RTP	8.10***	21.51***	3.12***	13.04***	-10.40***	-1.65***
	(1.21)	(1.31)	(0.26)	(1.86)	(0.70)	(0.14)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	896,999	3,408,709	3,408,709	896,999
Adjusted R^2 (%)	5.69	5.02	5.38	3.48	4.23	4.33
Panel B: Average flows over the next 5 days						
RTP	7.33***	21.75***	3.11***	11.31***	-11.81***	-1.61***
	(1.10)	(1.27)	(0.28)	(1.86)	(0.79)	(0.15)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	892,445	3,408,709	3,408,709	892,445
Adjusted R^2 (%)	4.43	6.76	9.22	6.32	8.50	7.95
Panel C: Average flows over the next 20 days						
RTP	6.85***	22.05***	3.13***	8.50***	-13.24***	-1.58***
	(1.04)	(1.25)	(0.31)	(1.72)	(0.85)	(0.17)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	875,366	3,408,709	3,408,709	875,366
Adjusted R^2 (%)	4.93	9.55	12.91	9.02	12.13	12.39

Table 10. Day trading proportion and trader characteristics

This table presents results from cross-sectional OLS regressions of day trading proportion on trader characteristics. The dependent variable is the percentage of total trading volume (in KRW) attributable to stock-days on which the investor both buys and sells the stock. Male is an indicator for male investors. Young is an indicator for investors aged 35 or below. High invest. know. is an indicator for investors who self-report the highest level of investment knowledge (4 on a scale of 1 to 4). MTS and HTS are indicators for investors whose main trading medium is mobile or desktop, respectively. Standard errors are reported in parentheses. The sample covers approximately 235,000 active retail accounts from January 2015 to December 2018, excluding accounts without a computable realized return and account holders younger than 20 or older than 85. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Day trade proportion (%)			
	(1)	(2)	(3)	(4)
Male	4.03***		2.94***	2.17***
	(0.09)		(0.09)	(0.11)
Young	3.94***		3.68***	1.23***
	(0.12)		(0.12)	(0.20)
High invest. know.	5.54***		5.19***	5.03***
	(0.14)		(0.14)	(0.25)
MTS		6.80***	6.06***	6.11***
		(0.11)	(0.11)	(0.11)
HTS		13.80***	13.05***	13.10***
		(0.13)	(0.13)	(0.13)
Male × Young				4.08***
				(0.25)
Male × High invest. know.				0.33
				(0.31)
Young × High invest. know.				-1.86***
				(0.62)
Male × Young × High invest. know.				2.21***
				(0.76)
Intercept	13.25***	10.31***	7.85***	8.27***
	(0.08)	(0.09)	(0.10)	(0.11)
N	235,445	235,445	235,445	235,445
Adjusted R^2 (%)	1.99	4.63	6.06	6.20

Table 11. Robustness to alternative mechanisms

This table presents results from Fama–MacBeth regressions of one-month-ahead returns on retail trading proportion (RTP) and other variables proposed in prior studies to explain overnight and intraday returns. Panel A reports one-month-ahead overnight returns and Panel B reports one-month-ahead intraday returns as dependent variables. Returns are expressed in percentage points. All columns include the control variables of Table 7. Detailed definitions of all variables are provided in Appendix Table A1. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Newey–West standard errors are reported in parentheses, computed with six lags. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Overnight return						
RTP	1.05***	1.03***	0.90***	0.97***	0.96***	0.94***
	(0.10)	(0.10)	(0.09)	(0.09)	(0.09)	(0.09)
Idiosyncratic volatility	0.43***	0.46***	0.43***	0.40***	0.39***	0.37***
	(0.07)	(0.08)	(0.07)	(0.07)	(0.07)	(0.07)
Abnormal volume		-0.10*	-0.08	-0.09*	-0.09*	-0.09*
		(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Overnight systematic variance			0.40***	0.39***	0.25	0.25
			(0.07)	(0.07)	(0.16)	(0.16)
Retail order imbalance (open)				0.25***	0.25***	0.25***
				(0.02)	(0.02)	(0.02)
Overnight beta					0.17	0.17
					(0.16)	(0.16)
Abnormal night reversal						-0.11***
						(0.02)
N	169,961	167,085	163,985	163,985	163,985	163,985
Adjusted R^2 (%)	12.63	12.94	13.40	13.53	13.75	13.79
Panel B: Intraday return						
RTP	-1.26***	-1.25***	-1.08***	-1.08***	-1.08***	-1.07***
	(0.11)	(0.11)	(0.10)	(0.09)	(0.09)	(0.09)
Idiosyncratic volatility	-1.09***	-1.08***	-1.00***	-1.00***	-1.00***	-0.97***
	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)
Abnormal volume		0.03	-0.02	-0.02	-0.02	-0.02
		(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Overnight systematic variance			-0.55***	-0.54***	-0.76***	-0.76***
			(0.10)	(0.10)	(0.15)	(0.15)
Retail order imbalance (open)				0.01	0.01	0.01
				(0.04)	(0.03)	(0.04)
Overnight beta					0.24*	0.25*
					(0.14)	(0.14)
Abnormal night reversal						0.10**
						(0.04)
N	169,961	167,085	163,985	163,985	163,985	163,985
Adjusted R^2 (%)	9.38	9.62	9.95	10.01	10.07	10.10

Table 12. Heterogeneous effects of RTP across subsamples

This table reports Fama–MacBeth regressions of one-month-ahead overnight returns (Panel A) and intraday returns (Panel B), in percentage points, on RTP, an indicator equal to one when a mechanism variable exceeds its cross-sectional median, and their interaction. Columns 1–6 include one mechanism variable at a time, and column 7 includes all six jointly. All columns include the control variables of Table 7, and Appendix Table A1 defines all variables. Independent variables are winsorized at the 1% and 99% levels and standardized within month. Newey–West standard errors with six lags are in parentheses. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Overnight return							
RTP	1.00***	1.02***	0.99***	0.96***	1.01***	1.08***	0.89***
	(0.09)	(0.10)	(0.09)	(0.11)	(0.09)	(0.10)	(0.10)
RTP × High indicator for:							
Idiosyncratic volatility	0.12***						0.15***
	(0.05)						(0.05)
Abnormal volume		0.03					-0.05
		(0.04)					(0.04)
Overnight systematic variance			-0.07				0.56
			(0.06)				(1.41)
Retail order imbalance (open)				0.21***			0.23***
				(0.07)			(0.07)
Overnight beta					-0.07		-0.66
					(0.06)		(1.40)
Abnormal night reversal						-0.08*	-0.05
						(0.04)	(0.04)
N	169,961	167,085	166,787	169,961	169,171	168,660	163,985
Adjusted R^2 (%)	12.74	12.75	12.83	12.79	12.92	12.67	13.22
Panel B: Intraday return							
RTP	-1.10***	-1.20***	-1.15***	-1.09***	-1.16***	-1.28***	-0.83***
	(0.10)	(0.12)	(0.09)	(0.13)	(0.09)	(0.12)	(0.11)
RTP × High indicator for:							
Idiosyncratic volatility	-0.43***						-0.40***
	(0.07)						(0.07)
Abnormal volume		-0.11					-0.01
		(0.07)					(0.07)
Overnight systematic variance			-0.10				-1.89
			(0.09)				(1.37)
Retail order imbalance (open)				-0.24***			-0.27***
				(0.07)			(0.07)
Overnight beta					-0.10		1.84
					(0.09)		(1.37)
Abnormal night reversal						0.05	0.04
						(0.05)	(0.05)
N	169,961	167,085	166,787	169,961	169,171	168,660	163,985
Adjusted R^2 (%)	9.49	9.53	9.57	9.47	9.58	9.41	9.75

Appendix A Variable definitions

Table A1. Variable definitions

This table defines all variables used in the analysis. Variables are computed using KRX data (January 2009 to December 2023) unless otherwise noted. Brokerage variables are from a large Korean discount broker (January 2015 to December 2018).

Variable	Definition
Panel A: Return variables	
Overnight return	Close-to-open return. For day s, it is computed as $r_{i,s}^{\text{overnight}} = (1 + r_{i,s})/(1 + r_{i,s}^{\text{intraday}}) - 1$, where $r_{i,s}$ is the close-to-close return and $r_{i,s}^{\text{intraday}}$ is the intraday return. The monthly level overnight return is computed as $r_{i,t}^{\text{overnight}} = \prod_{s \in \text{month } t}(1 + r_{i,s}^{\text{overnight}}) - 1$.
Intraday return	Open-to-close return. For day s, it is computed as $r_{i,s}^{\text{intraday}} = P_{i,s}^{\text{close}}/P_{i,s}^{\text{open}} - 1$, where $P_{i,s}^{\text{open}}$ and $P_{i,s}^{\text{close}}$ are the opening and closing auction prices. The monthly level intraday return is computed as $r_{i,t}^{\text{intraday}} = \prod_{s \in \text{month } t}(1 + r_{i,s}^{\text{intraday}}) - 1$
Return gap	Difference between overnight and intraday returns. For day s, it is computed as $\text{RG}_{i,s}^{d} = r_{i,s}^{\text{overnight}} - r_{i,s}^{\text{intraday}}$. The monthly return gap compounds the two legs: $\text{RG}_{i,t}^{m} = (1 + r_{i,t}^{\text{overnight}})/(1 + r_{i,t}^{\text{intraday}}) - 1$
Close-to-close return	Daily return from close to close: $r_{i,t} = P_{i,t}^{\text{close}}/P_{i,t-1}^{\text{close}} - 1$
Panel B: Trading activity variables	
RTP	Retail trading proportion, computed as the ratio of retail trading volume to total trading volume over a specified period (typically a day or a month)
Net buy	Buy volume minus sell volume, normalized by shares outstanding (in basis points)
Order imbalance (OIB)	Buy volume minus sell volume of an investor group, divided by the group's buy plus sell volume.
Panel C: Stock characteristics	
Log market equity	Natural logarithm of market capitalization (stock price $\times$ shares outstanding), measured at the end of month t, immediately before the return month
Book-to-market	Book equity divided by market equity, measured at the beginning of the month
Market beta	Estimated using past 120 trading days of returns, with the KOSPI index as the market portfolio
Amihud illiquidity	Average of $\lvert\text{ret}_{i,t}\rvert/\text{volume}_{i,t}$ over the past 120 trading days, where volume is measured in KRW billions
Foreign ownership	Percentage of shares outstanding held by foreign investors, measured at the beginning of the month
Past 120-day return	Cumulative return over the past 120 trading days

Table A1. Variable definitions (continued)

Variable	Definition
Panel C: Stock characteristics (continued)	
Idiosyncratic volatility	Standard deviation of residuals in a regression of individual stock return on Fama-French three factors in Korean market, using daily data in the current month (over the trailing 21 trading days in the daily regressions of Table 9)
Maximum daily return	Maximum daily return over the trading days in the current month (over the trailing 21 trading days in the daily regressions of Table 9)
Panel D: Alternative mechanism variables	
Abnormal volume	Difference between month t turnover and average turnover over the prior 12 months
Overnight systematic variance	Squared overnight beta times market overnight return variance, where overnight beta is estimated using a rolling 12-month window with at least 100 daily observations
Retail order imbalance (open)	Aggressive retail buy-minus-sell volume during the first trading hour, scaled by twice the daily volume, aggregated monthly
Overnight beta	Estimated from a rolling regression of overnight stock returns on overnight market returns over 12 months with at least 30 observations
Abnormal night reversal	Current-month proportion of days with positive overnight and negative intraday returns, divided by its average over the trailing twelve months including the current month
Panel E: Brokerage account variables	
Male	Binary indicator equal to 1 if the account holder is male
Age	Account holder's age in years at the end of 2018
Average daily balance	Mean daily account balance over the sample period (in KRW millions)
Average trade size	Mean trade size over the sample period (in KRW millions)
Number of unique stocks	Total number of unique stocks traded during the sample period
Number of trades	Total number of trades executed during the sample period
Performance	Realized return over the sample period, winsorized at 1% and 99%
Main trading medium	Trading interface (MTS, HTS, or Other) through which the trader executed the most trades
Panel F: Brokerage questionnaire variables	
Monthly income	Self-reported monthly income: 1 = less than KRW 1M, 2 = KRW 1–3M, 3 = KRW 3–6M, 4 = KRW 6–10M, 5 = more than KRW 10M
Investment knowledge	Self-reported investment knowledge: 1 = little understanding, 2 = partial understanding, 3 = deep understanding, 4 = understanding of most products including derivatives
Investment aggressiveness	Self-reported risky product experience: 1 = none, 2 = government bonds/MMFs, 3 = bond funds/high-credit bonds, 4 = mixed funds/medium-credit bonds, 5 = stocks/stock funds, 6 = derivatives/leveraged products
Derivative experience	Self-reported derivative trading experience: 1 = less than 1 year, 2 = 1–3 years, 3 = more than 3 years

Appendix B The opening and closing auctions

Here, we take a more careful look at the opening and closing auctions. Because the return gap in our main analysis is defined using the opening and closing auction clearing prices, a natural concern is that it reflects the peculiarities of these auctions rather than the retail demand behind them. This appendix demonstrates that this is not the case. Our results are unaffected by the choice of measuring prices and flows at the auctions alone or over the thirty-minute windows at the market open and close.

Figure A1 decomposes close-to-close returns into four segments. On average, prices rise 10.2 basis points overnight, from the previous day's closing auction to the opening auction. They then fall 3.9 basis points from the opening auction to the volume-weighted average price (VWAP) of the first thirty minutes of continuous trading, and another 12.0 basis points between the first and last thirty-minute VWAPs. Both VWAPs exclude auction trades. Prices recover 8.3 basis points from the last thirty-minute VWAP into the closing auction. This recovery may reflect the greater institutional participation in the closing auction (Table A3).

[Figure A1 here]

Figure A2 shows that the return differences associated with auction RTP extend beyond the auction prices to the surrounding continuous-trading windows. Each day, we sort stocks into terciles of the prior-month average retail trading proportion (RTP) in the corresponding auction, calculated as retail buy plus sell volume divided by twice the auction volume.[25]

Stocks with high opening-auction RTP earn higher overnight returns than those with low RTP, and the ordering of the terciles persists at every subsequent VWAP through 11:00. Near the close, stocks with high closing-auction RTP have lower cumulative returns than those with low RTP, with the high-minus-low differential widest in the final interval of continuous trading before the closing auction. Prices then rise in the closing auction for all three terciles, narrowing the differential without reversing its sign.

[Figure A2 here]

In Figure A3, we turn from prices to flows. We plot retail net buy volume during the first and last thirty minutes of the trading day and separate the flow executed in the auctions from the flow

[25] We use the prior-month average because day-level shares leave little cross-sectional variation. The median stock-day opening auction is 96% retail by volume.

executed in continuous trading. Again, the pattern is consistent within and around the auctions. Retail investors are net buyers at the open and net sellers at the close, in the auctions and the adjacent continuous windows alike (Panel (a), which normalizes by retail volume in each segment). The auctions themselves are modest slices of trading at the two ends of the day. The opening auction accounts for 8.7% of volume in the first thirty minutes, and the closing auction for 33.5% in the last thirty minutes. Panel (b) accordingly shows that, relative to shares outstanding as in Figure 3, the auction imbalances are far smaller than the continuous-trading flows (an order of magnitude at the open). Normalizing by retail volume in Panel (a) is what makes the auction tilts visible.

[Figure A3 here]

In Table A3, we further compare retail participation measured within the auctions with the monthly RTP used throughout the paper, to verify that RTP also represents what happens during and around the auctions. The within-auction measures align closely with RTP. Within months, the cross-sectional correlation between monthly RTP and RTP within the auctions is 0.85 at the open and 0.90 at the close, and it rises to 0.94 and 0.97 for the surrounding thirty-minute continuous-trading windows that exclude the auctions.

[Table A3 here]

Table A4 relates RTP in each auction to the corresponding return leg. Higher opening-auction RTP is associated with higher overnight returns, while higher closing-auction RTP is associated with lower intraday returns. These patterns hold using both same-day auction RTP and its prior-month average.

[Table A4 here]

Lastly, our results are unaffected when we repeat the paper's Fama–MacBeth regressions with overnight and intraday returns computed from the first and last thirty-minute VWAPs instead of the auction prices (Table A2). Together, the price and flow evidence indicates that the return gap is not an artifact of the auction mechanism. The auction prices are consistent with the prices around them, and retail flow tilts the same way inside and outside the auctions.

[Table A2 here]

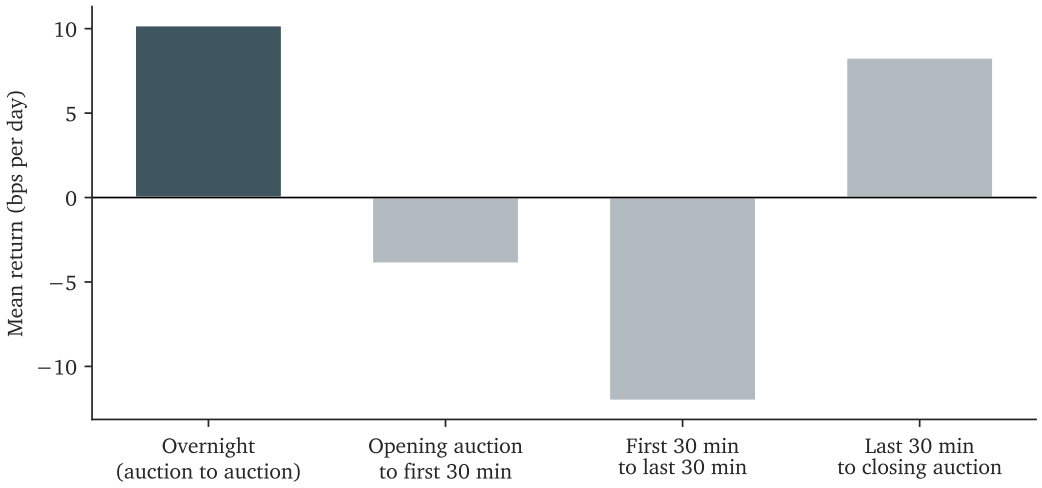
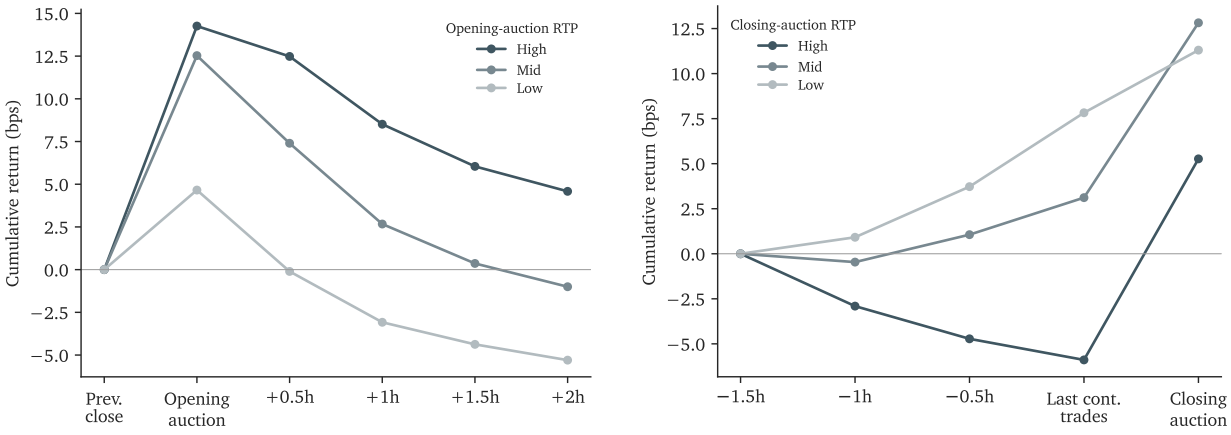


Figure A1. Return decomposition at the opening and closing auctions

This figure decomposes daily close-to-close returns at the auction boundaries and reports equal-weighted mean returns per day in basis points. The overnight segment runs from the previous day's closing auction price to the opening auction price, adjusted for corporate actions. The three intraday segments are computed from same-day prices, where the first (last) thirty-minute VWAP covers continuous-session trades during the first (last) thirty minutes of the trading day, excluding the auction trades.

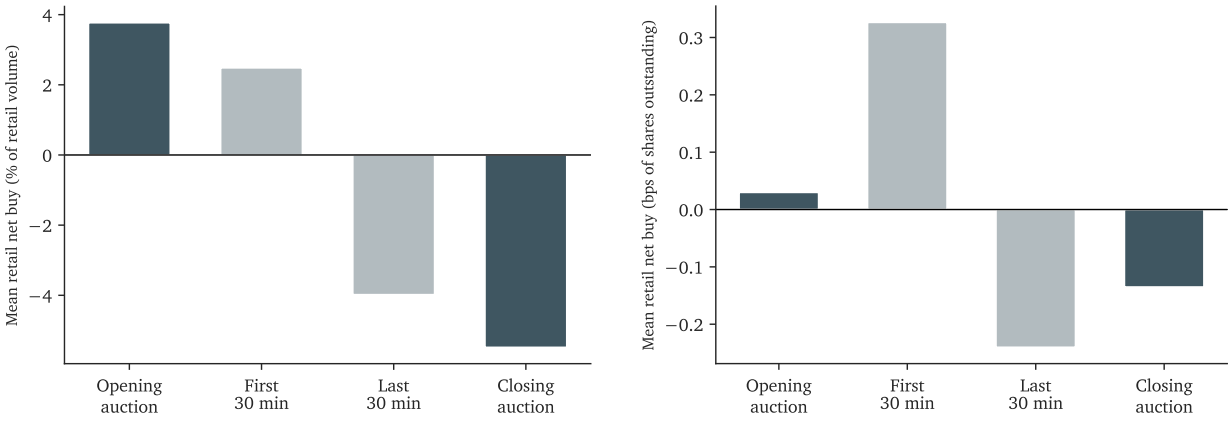


(a) Around the opening auction

(b) Into the closing auction

Figure A2. Price paths around the auctions by auction retail participation

This figure plots equal-weighted average cumulative returns in basis points around the opening auction (Panel (a)) and into the closing auction (Panel (b)), for daily terciles of the prior-month average RTP of the corresponding auction. In Panel (a), paths are measured relative to the previous day's closing auction price. The overnight step uses daily returns adjusted for corporate actions, and the remaining points are thirty-minute VWAPs of continuous trading ending half an hour to two hours after the open (9:30 to 11:00). In Panel (b), paths are measured relative to the thirty-minute VWAP ending ninety minutes before the close, followed by two thirty-minute VWAPs, the VWAP of the final continuous trades, and the closing auction price. Points are aligned by time to the close, so the alignment is unaffected by the August 2016 extension of the trading session. For each plotted point, we first average cumulative returns equally across stocks within each day and tercile, then average these daily tercile means equally across days. The sample period is March 2009, when the exchange's dedicated session identifier enters the trade records, to December 2023.



(a) Relative to retail volume

(b) Relative to shares outstanding

Figure A3. Retail order flow at the auctions and the adjacent windows

This figure plots mean retail net buy volume for the opening auction, the first thirty minutes of continuous trading, the last thirty minutes of continuous trading, and the closing auction. Panel (a) normalizes by retail buy plus sell volume in the same segment, in percent, excluding stock-days with no retail volume in the segment. Panel (b) normalizes by shares outstanding, in basis points, as in Figure 3. Averages are equal-weighted across stock-days. The sample covers 3,546,506 stock-days with matched opening and closing auction trades from March 2009, when the exchange's dedicated session identifier enters the trade records, to December 2023.

Table A2. Fama–MacBeth regressions using VWAP-based returns

This table presents results from Fama–MacBeth regressions using returns computed from volume-weighted average prices (VWAPs). The dependent variables are one-month-ahead overnight returns in columns 1–3, one-month-ahead intraday returns in columns 4–6, and one-month-ahead return gaps in columns 7–9. Overnight and intraday returns are computed using VWAPs during the first and last 30 minutes of each trading day (market open and close), respectively. Returns are expressed in percentage points. The key independent variable is RTP. Detailed variable definitions are provided in Appendix Table A1. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Newey–West standard errors are reported in parentheses, computed with six lags. Industry fixed effects include 51 KRX industry classifications. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Overnight return			Intraday return			Return gap		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
RTP	2.17***	1.91***	1.90***	-2.13***	-2.15***	-2.16***	4.85***	4.19***	4.17***
	(0.15)	(0.10)	(0.10)	(0.12)	(0.11)	(0.11)	(0.27)	(0.18)	(0.18)
Overnight return		0.91***	0.86***		-0.89***	-0.91***		2.09***	2.06***
		(0.09)	(0.09)		(0.07)	(0.08)		(0.19)	(0.20)
Intraday return		-0.73***	-0.79***		0.65***	0.58***		-1.47***	-1.44***
		(0.06)	(0.06)		(0.07)	(0.07)		(0.13)	(0.13)
Log market equity		0.48***	0.64***		-0.87***	-0.99***		1.26***	1.57***
		(0.09)	(0.07)		(0.08)	(0.09)		(0.13)	(0.12)
Market beta		0.54***	0.50***		-0.41***	-0.39***		1.01***	0.94***
		(0.07)	(0.07)		(0.06)	(0.06)		(0.12)	(0.12)
Book-to-market		-0.21***	-0.06		0.24***	0.22***		-0.54***	-0.37***
		(0.07)	(0.06)		(0.07)	(0.06)		(0.10)	(0.10)
Amihud illiquidity		0.02	0.00		0.11**	0.09*		-0.10	-0.09
		(0.04)	(0.04)		(0.05)	(0.05)		(0.09)	(0.09)
Foreign ownership		0.12***	0.09**		-0.09**	-0.12***		0.18***	0.17***
		(0.04)	(0.04)		(0.04)	(0.04)		(0.06)	(0.06)
Past 120-day return		0.21***	0.16**		-0.05	-0.05		0.35**	0.30**
		(0.07)	(0.07)		(0.09)	(0.09)		(0.15)	(0.15)
Idiosyncratic volatility		-0.02	0.02		-0.64***	-0.63***		1.25***	1.27***
		(0.09)	(0.09)		(0.09)	(0.09)		(0.17)	(0.18)
Maximum daily return		0.37***	0.41***		-0.03	-0.01		0.44***	0.45***
		(0.07)	(0.07)		(0.08)	(0.09)		(0.14)	(0.15)
Industry FE			✓			✓			✓
N	166,927	166,927	156,737	166,927	166,927	156,737	166,927	166,927	156,737
Adjusted R^2 (%)	6.40	13.39	17.37	5.89	13.50	16.68	9.05	18.08	20.68

Table A3. RTP during the opening and closing auctions

This table compares the retail trading proportion (RTP) measured within the opening and closing call auctions and in the adjacent thirty-minute windows with the paper's monthly RTP. RTP within an auction is retail buy plus sell volume divided by twice the matched auction volume, that is, the paper's RTP computed within the auction. First (last) 30 min excl. auction is RTP computed over continuous-session trades during the first (last) thirty minutes of the trading day, excluding the auction trades. All measures are aggregated to the stock-month level. Panel A reports summary statistics pooled across stock-months. Panel B reports the time-series average of monthly cross-sectional Pearson correlations. The sample period is March 2009, when the exchange's dedicated session identifier enters the trade records, to December 2023.

Panel A: Summary statistics

	Mean	SD	P25	P50	P75
(1) RTP	0.73	0.20	0.61	0.78	0.89
(2) Opening auction, RTP	0.87	0.17	0.82	0.93	0.98
(3) First 30 min excl. auction	0.77	0.17	0.68	0.80	0.90
(4) Closing auction, RTP	0.73	0.27	0.57	0.84	0.94
(5) Last 30 min excl. auction	0.71	0.22	0.57	0.78	0.89

Panel B: Within-month cross-sectional correlations

	(1)	(2)	(3)	(4)	
(1) RTP					
(2) Opening auction, RTP	0.85				
(3) First 30 min excl. auction	0.94	0.86			
(4) Closing auction, RTP	0.90	0.84	0.85		
(5) Last 30 min excl. auction	0.97	0.82	0.90	0.89	

Table A4. Auction RTP and the two return legs

This table reports daily Fama–MacBeth regressions of overnight returns, in columns 1 and 2, and intraday returns, in columns 3 and 4, on the RTP of the auction that sets each leg's terminal price: the opening auction for overnight returns and the closing auction for intraday returns. The auction RTP is retail buy plus sell volume divided by twice the matched auction volume, as in Table A3, entering either on the same day, in columns 1 and 3, or as its average over the prior month, requiring at least five daily observations, in columns 2 and 4. Returns are in basis points, and the regressors are winsorized at the 1st and 99th percentiles and standardized within each day, so coefficients measure basis points per one within-day standard deviation. *t*-statistics computed from Newey–West standard errors with five lags on the daily coefficient series appear in brackets. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is March 2009, when the exchange's dedicated session identifier enters the trade records, to December 2023.

	Overnight return		Intraday return	
	(1)	(2)	(3)	(4)
Opening-auction RTP, same day	3.83***			
	[12.93]			
Opening-auction RTP, prior-month average		4.84***		
		[16.85]		
Closing-auction RTP, same day			-6.63***	
			[-14.01]	
Closing-auction RTP, prior-month average				-4.51***
				[-9.39]
N	3,553,949	3,513,594	3,519,628	3,511,130
Days	3,652	3,645	3,652	3,645

Appendix C The nominal-price instrument

In this appendix, we explore the robustness of the functional form of our nominal-price instrument and potential explanations for the magnitude difference between the ordinary least squares (OLS) and two-stage least squares (2SLS) estimates. We provide two additional explanations beyond the one discussed in the main text, namely that sophisticated investors partially trade against predictable price pressure and thereby attenuate the reduced-form OLS estimate.

C.1 Functional form

In Table A5, we compare alternative instrument sets constructed from nominal price: log price, a quadratic in log price, a piecewise-linear spline in log price, and nominal-price-bin indicators. The log-price instrument delivers the strongest first stage (a Kleibergen–Paap (KP) Wald F of 318.4, against 192.4, 83.8, and 64.5 for the alternatives), and the second-stage return-gap estimates are stable across instrument sets, from 4.37 to 4.58. Given these results, we choose the just-identified log-price specification, in line with the standard preference for a small number of strong instruments (e.g., Stock and Yogo 2005, Angrist and Pischke 2009).

[Table A5 here]

C.2 The local average treatment effect and compliers

Because the instrument shifts retail participation through nominal-price accessibility, the 2SLS coefficient is a local average treatment effect (LATE): it averages treatment effects over stocks whose retail participation responds to nominal price. The brokerage evidence in Figure 7 suggests that this response operates through lower-balance investors, who favor low-priced stocks and also day-trade more. The estimate may therefore place greater weight on stocks whose retail clientele has a stronger preference for daytime exposure. If participation by these investors generates greater price pressure, this weighting could explain the larger 2SLS estimate.

C.3 Persistence and the OLS–2SLS gap

Another potential contributor to the larger 2SLS estimate is the persistence of the component of RTP that the instrument isolates. In our main specification, we regress returns on lagged RTP to mitigate simultaneity concerns. We show below that a gap between the two estimators arises in this design

when the true relationship is contemporaneous. OLS on lagged RTP loads on transitory variation in retail participation that does not carry forward, while the instrument isolates the persistent component that does.

Setup. Suppose the true relationship is contemporaneous:

$$y_{i,t} = \beta \, \text{RTP}_{i,t} + \varepsilon_{i,t} \tag{A1}$$

where $y_{i,t}$ is the overnight or intraday return. Assume RTP has two components:

$$\text{RTP}_{i,t} = \text{RTP}^s_{i,t} + \text{RTP}^f_{i,t} \tag{A2}$$

where RTP^s is a slow-moving (persistent) component and RTP^f is a fast-moving (transitory) component. Specifically:

$$\text{RTP}^s_{i,t} = \rho_s \, \text{RTP}^s_{i,t-1} + u^s_{i,t}, \quad \rho_s < 1 \tag{A3}$$

$$\text{RTP}^f_{i,t} = \rho_f \, \text{RTP}^f_{i,t-1} + u^f_{i,t}, \quad \rho_f \approx 0 \tag{A4}$$

Note $u^s_{i,t}$ and $u^f_{i,t}$ are independent shocks. The instrument $z_{i,t-1}$ (lagged log price) is correlated with the slow-moving component $\text{RTP}^s_{i,t-1}$ but uncorrelated with the transitory component $\text{RTP}^f_{i,t-1}$.

OLS with lagged RTP. When we regress $y_{i,t+1}$ on $\text{RTP}_{i,t}$, we can substitute the true model:

$$y_{i,t+1} = \gamma \, \text{RTP}_{i,t} + \varepsilon_{i,t+1} = \gamma \left(\text{RTP}^s_{i,t} + \text{RTP}^f_{i,t} \right) + \varepsilon_{i,t+1} \tag{A5}$$

Note the slow-moving component of RTP is highly persistent, i.e., $\text{RTP}^s_{i,t+1} \approx \text{RTP}^s_{i,t}$, and the fast moving component is largely uncorrelated over time, i.e., $\text{cov}(\text{RTP}^f_{i,t+1}, \text{RTP}^f_{i,t}) \approx 0$. Therefore, the coefficient on lagged RTP captures primarily the slow-moving component. The OLS coefficient converges to:

$$\hat{\gamma}_{\text{OLS}} \xrightarrow{p} \beta \left(\rho_s \omega_s + \rho_f \omega_f \right) \tag{A6}$$

where $\omega_s = \text{var}(\text{RTP}^s_{i,t})/\text{var}(\text{RTP}_{i,t})$ and $\omega_f = \text{var}(\text{RTP}^f_{i,t})/\text{var}(\text{RTP}_{i,t})$ are the variance shares of the slow and fast components. Because $\rho_f \approx 0$, the OLS coefficient is attenuated relative to β as $\rho_s < 1, \omega_s < 1$.

2SLS with lagged RTP. The instrument $z_{i,t}$ isolates variation in $\text{RTP}^s_{i,t}$, which persists into period $t+1$. The 2SLS coefficient converges to:

$$\begin{aligned}\hat{\gamma}_{\text{2SLS}} \xrightarrow{p} \frac{\text{cov}(y_{i,t+1}, z_{i,t})}{\text{cov}(\text{RTP}_{i,t}, z_{i,t})} &= \frac{\text{cov}(\beta(\text{RTP}^s_{i,t+1} + \text{RTP}^f_{i,t+1}), z_{i,t})}{\text{cov}(\text{RTP}^s_{i,t} + \text{RTP}^f_{i,t}, z_{i,t})} \\ &= \frac{\beta\rho_s \text{cov}(\text{RTP}^s_{i,t}, z_{i,t})}{\text{cov}(\text{RTP}^s_{i,t}, z_{i,t})} = \beta\rho_s \approx \beta \end{aligned} \tag{A7}$$

The second equality holds due to $\text{cov}(\text{RTP}^f_{i,t}, z_{i,t}) = 0$ and $\text{cov}(u^s_{i,t+1}, z_{i,t}) = 0$ (instrument uncorrelated with future shocks in slow-moving component). Since the instrument captures only the persistent component, 2SLS recovers a coefficient closer to β, while OLS is attenuated by the transitory component that does not carry forward.

Table A5. Alternative instrument specifications for RTP

This table compares four instrument specifications for RTP in the two-stage least squares estimation of Table 8. The first is the log of the nominal share price on the first trading day of the month. The second adds the squared demeaned log price. The third is a linear spline in log price with knots at the KRX tick-size brackets (KRW 5,000, 10,000, 50,000, and 100,000). The fourth is a set of indicator variables for nominal-price bins with cut points at KRW 1,000, 5,000, 10,000, 50,000, 100,000, and 500,000, omitting the most populated bin. Each row reports the number of instruments, the Kleibergen–Paap (KP) rank Wald F statistic as in Table 8, the first-stage partial R^2, and the second-stage coefficient on RTP for one-month-ahead overnight returns, intraday returns, and return gaps (in percent). The last row reports the corresponding OLS estimates without an instrument. All specifications include the full control set of Table 7, month and industry fixed effects, and standard errors double-clustered by firm and month, with RTP and controls winsorized at the 1st and 99th percentiles and standardized within month. Standard errors are in parentheses; ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is January 2009 to December 2023.

	# IVs	KP F	Partial R^2 (%)	Overnight	Intraday	Return gap
ln(price)	1	318.4	10.03	2.28***	-1.90***	4.51***
				(0.24)	(0.30)	(0.45)
ln(price) + ln(price)^2	2	192.4	10.07	2.32***	-1.93***	4.58***
				(0.24)	(0.30)	(0.45)
Linear spline in ln(price)	5	83.8	11.05	2.20***	-1.85***	4.38***
				(0.23)	(0.28)	(0.43)
Price-bin dummies	6	64.5	8.38	2.23***	-1.77***	4.37***
				(0.25)	(0.30)	(0.47)
OLS (no instrument)	0			0.99***	-1.24***	2.36***
				(0.09)	(0.11)	(0.16)
N					159,472	

Appendix D Robustness across samples

This appendix demonstrates the robustness of the paper's main results across samples. Table A6 reports the Fama–MacBeth and 2SLS estimates of the RTP coefficient for the paper's baseline larger half of the cross-section, the full universe including the smaller half, the two exchanges (KOSPI and KOSDAQ) separately, and three subperiods split at the June 2015 widening of daily price limits and the early-2020 onset of the pandemic. We also include a sample that adds cash dividends back to the ex-date overnight returns. Each reported coefficient comes from a separate Fama–MacBeth or 2SLS regression with the same set of controls as in Table 7. In every sample we examine, the RTP coefficients keep their signs, and all remain statistically significant except the 2SLS intraday estimate in the final subperiod. We do not report estimates for the smaller half alone because RTP has limited cross-sectional variation among these stocks. RTP averages 0.91 in this group, with an average within-month standard deviation of 0.06, compared with 0.20 in the larger half.

[Table A6 here]

One noticeable pattern is that the return-gap estimates decline across the three subperiods, although they remain positive and statistically significant. The decline is consistent with the growth of professional short-horizon liquidity provision over the sample. The trading proportions of short-term institutions and of foreign investors in the average stock-month roughly double from the first subperiod to the second and rise further in the third. Consistent with this reading, the interactions of RTP with above-median short-term institution and foreign trading proportions in Table A9 each attenuate the return-gap effect by about a quarter.

Table A6. Robustness of the main estimates

This table reports the coefficient on RTP from regressions of one-month-ahead overnight returns, intraday returns, and return gaps (in percent) on RTP across samples. Each cell is a separate regression. OLS columns are Fama–MacBeth regressions as in Table 7, with the full control set, industry fixed effects, and Newey–West standard errors with six lags. 2SLS columns instrument RTP with the logarithm of the nominal share price as in Table 8, with month and industry fixed effects and standard errors double-clustered by firm and month. RTP and all controls are winsorized at the 1st and 99th percentiles and standardized within month and sample, so coefficients measure percent per one within-month standard deviation. The baseline sample is the larger half of stocks by beginning-of-year market capitalization from January 2009 to December 2023. All stocks adds the smaller half. The exchange rows split the baseline sample by the stock's listing venue, and the subperiod rows split it at the June 2015 widening of daily price limits from 15% to 30% and at the onset of the pandemic in early 2020. The dividend-adjusted row repeats the baseline sample after adding each cash dividend back to the ex-date overnight return, which leaves intraday returns unchanged. The Kleibergen–Paap F statistic for the instrument exceeds 150 in every sample. Standard errors are in parentheses; ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Overnight return		Intraday return		Return gap		
	OLS	2SLS	OLS	2SLS	OLS	2SLS	N
Baseline: Larger half	1.00***	2.28***	-1.22***	-1.90***	2.35***	4.51***	159,472
	(0.09)	(0.24)	(0.11)	(0.30)	(0.18)	(0.45)	
All stocks	0.66***	3.25***	-1.06***	-2.23***	1.75***	6.26***	296,616
	(0.08)	(0.32)	(0.08)	(0.36)	(0.14)	(0.59)	
KOSPI	1.09***	2.07***	-1.29***	-1.85***	2.46***	4.06***	87,368
	(0.11)	(0.26)	(0.13)	(0.31)	(0.21)	(0.48)	
KOSDAQ	0.56***	1.85***	-0.75***	-1.73***	1.48***	4.08***	72,104
	(0.06)	(0.26)	(0.09)	(0.36)	(0.13)	(0.55)	
Jan 2009–Jun 2015	1.24***	3.23***	-1.70***	-3.11***	3.10***	6.45***	57,208
	(0.11)	(0.37)	(0.11)	(0.45)	(0.19)	(0.66)	
Jul 2015–Feb 2020	1.03***	1.94***	-0.83***	-1.11***	2.01***	3.35***	51,405
	(0.10)	(0.28)	(0.13)	(0.42)	(0.18)	(0.58)	
Mar 2020–Dec 2023	0.55***	0.73*	-0.90***	-0.62	1.48***	1.95**	50,859
	(0.17)	(0.43)	(0.20)	(0.66)	(0.33)	(0.87)	
Dividend-adjusted returns	0.98***	2.26***	-1.22***	-1.90***	2.34***	4.48***	159,472
	(0.09)	(0.24)	(0.11)	(0.30)	(0.18)	(0.45)	

Appendix E Clientele effects

Prior work has hypothesized clientele effects behind overnight and intraday return patterns, in which distinct groups of investors concentrate their demands at different times of the trading day and the recurring imbalances between them move prices (Lou et al. 2019, 2026). The proposed sources of these recurring demands are largely institutional, such as funds rebalancing against the closing price and arbitrageurs unwilling to carry positions overnight (Bogousslavsky 2021, Bogousslavsky and Muravyev 2023). This appendix places the retail ebb and flow in this literature: a retail clientele whose origin is a preference for daytime exposure rather than an institutional constraint.

Owing to data constraints, most prior studies infer the activities of the clienteles from return patterns (e.g., Akbas et al. 2022). Exploiting trade records that identify the investor types on both sides of each transaction, we document two findings relevant to this literature and our work. First, the conjectured clientele effect exists: the time-of-day demands are persistent at the stock level, with retail investors exerting buying pressure in the morning and long-term institutions (non-retail institutions other than financial investment companies) in the afternoon on average. A separate arbitrageur-like group, short-term institutions (financial investment companies) and foreign investors, accommodates both. Second, the presence of this last group mitigates the price effects of retail demand.

Table A7 provides context on the trading horizons of different Korea Exchange (KRX) investor types by reporting their *intraday offset share*, the fraction of each type's trading volume offset within the same stock and day. Letting B and S denote the type's buy and sell volume for a given stock and day, we define this share as

$$\text{Intraday offset share} = \frac{2\min(B,S)}{B+S}.$$

For example, if an investor type buys 100 shares and sells 60 shares, 60 shares on each side offset, accounting for 120 of the 160 shares of gross trading volume, or 75%. The remaining 40 shares represent net purchases. Higher values indicate greater intraday turnover relative to the type's net position change, consistent with shorter trading horizons.[26]

Table A7 reports the volume-weighted average of this measure across stock-days. Financial investment companies, our short-term institutions, offset 71% of their volume within the day, against

[26]The offsetting buys and sells need not come from the same investors, so the measure may also reflect differences in beliefs or trading motives within an investor type.

roughly a third for insurance companies, private equity funds, and pension funds. Mobile and desktop retail investors have the highest offset shares, at over 90%.

[Table A7 here]

We then verify that the time-of-day demands are persistent at the stock level. For each stock and week, we compute net purchases by retail investors during the morning half of the trading day and by long-term institutions during the afternoon half, each scaled by the stock-week's total trading value. Within each week, we sort stocks into quintiles of one measure and track it over the following weeks. Table A8 shows strong persistence in both sorts.

[Table A8 here]

We next ask which side of each trade demands immediacy. Each executed trade records when the two matched orders arrived, so the initiator is identifiable. Figure A4 plots each group's net initiated volume as a share of interval volume, demeaned by the group's own all-day average, so the lines show when a group departs from its habitual order style. Retail investors, for instance, buy predominantly with resting limit orders and sell with marketable ones (Linnainmaa 2010, Kelley and Tetlock 2013). Given such stable order styles, the demeaned measure is the more informative one. Retail investors demand buy-side immediacy in the opening half hour and sell-side immediacy at the close. This rotation is what the preference for daytime exposure implies, and its behavioral motives are the subject of Section 4.2. Long-term institutions run the mirror image: they sell through midday and demand buy-side immediacy in the final half hour, when funds benchmarked to the closing price rebalance (Bogousslavsky and Muravyev 2023). Short-term institutions and foreign investors behave instead like liquidity suppliers: they sell into the elevated open and turn to buy toward the close, and we treat them as a single accommodating class below. Foreign investors in Korea are predominantly global institutions (Choe et al. 1999), the investor type that Grinblatt and Keloharju (2000) find to be the most sophisticated.

[Figure A4 here]

Table A9 relates the gap to the presence of these investors and to their role as retail's counterparties. In columns 2 to 4, we interact RTP with an indicator for an above-median trading proportion of short-term institutions and with indicators for above-median foreign ownership and foreign trading proportion, the two groups that Figure A4 shows consistently taking the other side of retail's

timed demand. Each presence interaction attenuates the return-gap effect by about a quarter of the corresponding RTP coefficient, consistent with short-horizon and foreign institutional capital leaning against retail-induced price pressure. Entered jointly, the attenuation loads on the two foreign measures (column 6).

[Table A9 here]

Finally, we test whether the composition of the other side matters for the gap. For each stock and month, we measure the *absorption share*, the fraction of retail buy volume in the opening half hour executed against foreign or short-term institutional sellers. For instance, if retail investors buy KRW 100 million of a stock in the first thirty minutes and KRW 20 million of it executes against these two groups, the absorption share is 20%. We measure absorption at the open because short-term institutions and foreign investors are retail's main professional counterparty there, whereas at the close much of retail's selling is met by long-term institutions' own demand (Figure A4). Entered alone, the absorption interaction lowers the gap's sensitivity to RTP by about a quarter, as the two trading-proportion interactions do (column 5). Entered together with them, it keeps its size while the two trading-proportion interactions become indistinguishable from zero (column 7). The attenuation thus reflects who takes the other side of retail's morning demand, not how much these investors trade in the stock overall. The gap is largest where the professional side of the market is thinnest.

Table A7. Trading horizon by investor category

This table reports, for each KRX investor category, its share of total trading volume and the share of its own volume offset within the same stock and day, defined as $2\min(B,S)/(B+S)$ aggregated across stock-days, where B and S are the category's buy and sell volume in a stock and day. The classification column shows the paper's grouping: financial investment companies are short-term institutions, all other non-retail categories are long-term institutions, and individuals are retail investors, split by order channel. Foreign investors are not a separate category in these records, as their orders route through the domestic type codes. Government and unclassified is a residual category that the exchange merged into pension funds from December 2018. The sample covers the baseline large-cap universe from January 2009 to December 2023.

KRX category	Classification	Volume share (%)	Offset within day (%)
Financial investment companies	Short-term institutions	7.0	70.8
Insurance companies	Long-term institutions	0.5	34.1
Investment trust companies	Long-term institutions	2.3	54.7
Private equity funds	Long-term institutions	0.5	27.6
Banks	Long-term institutions	1.0	51.1
Other financial companies	Long-term institutions	1.2	61.7
Pension funds	Long-term institutions	1.0	33.5
Government and unclassified	Long-term institutions	0.8	76.5
Other corporations	Long-term institutions	1.9	58.3
Individuals, mobile (MTS)	Retail investors	31.8	91.7
Individuals, home trading system (HTS)	Retail investors	46.6	94.4
Individuals, other channels	Retail investors	5.6	72.6

Table A8. Persistence of time-of-day flows

This table sorts stock-weeks into quintiles of net purchases by an investor clientele during its half of the trading day and reports the mean of the sorting measure in the formation week and 1, 4, and 12 weeks later. Panel A sorts on retail net purchases during the morning half of the trading day (9:00 to 12:00), and Panel B on long-term-institution net purchases during the afternoon half. Auction trades count toward the half in which they occur, and each measure is scaled by the stock-week's total buy plus sell value and expressed in percent. Quintiles are formed within each week among the larger half of stocks by beginning-of-year market capitalization. Long-term institutions are domestic non-retail investors other than financial investment companies. Foreign investors are identified by the foreign-investor code and separated from the domestic groups. The sample runs from January 2009 to December 2023.

	Week t	$t+1$	$t+4$	$t+12$
Panel A: Retail net purchases, morning half of the day				
Q1 (lowest)	-3.84	-0.97	-0.25	0.05
Q2	-0.88	-0.05	0.10	0.19
Q3	0.19	0.16	0.17	0.16
Q4	1.33	0.51	0.35	0.28
Q5 (highest)	4.44	1.58	0.86	0.56
Panel B: Long-term-institution net purchases, afternoon half of the day				
Q1 (lowest)	-4.36	-1.45	-0.51	-0.12
Q2	-0.82	-0.27	-0.08	0.00
Q3	0.01	0.01	0.02	0.04
Q4	1.01	0.45	0.26	0.19
Q5 (highest)	5.00	2.10	1.14	0.70

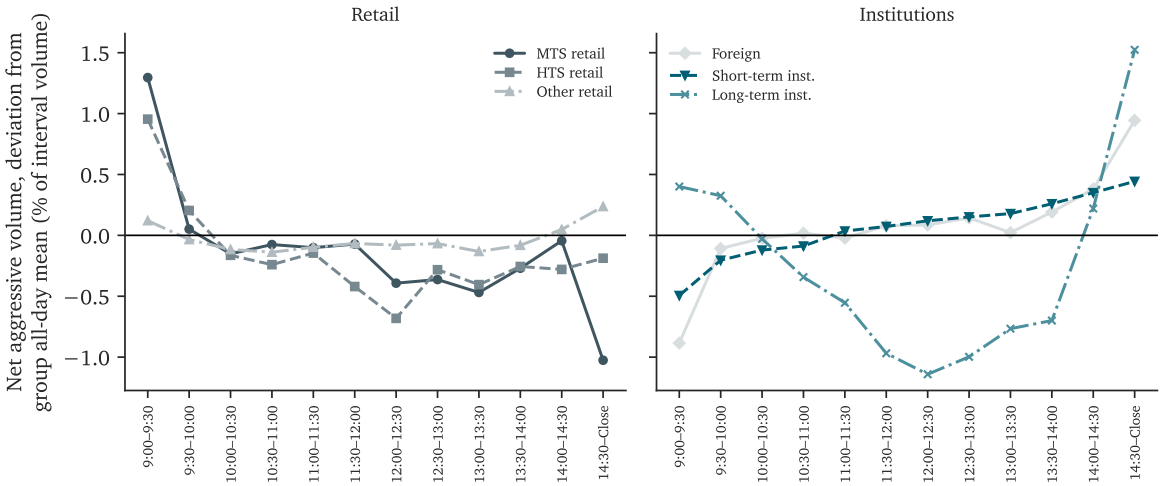


Figure A4. Net aggressive volume by investor group

This figure plots each investor group's net aggressive volume, buy-initiated minus sell-initiated value, as a share of total trading value in each 30-minute interval, demeaned by the group's own all-day average. The initiating side of a trade is the side whose order arrived later. Trades in which the two orders carry the same arrival time are excluded from initiated volume. Investor groups follow Section 2.2, with foreign investors identified by the foreign-investor code on either side of the trade and separated from the domestic groups. The sample covers continuous-session trades in the larger half of stocks by beginning-of-year market capitalization from January 2009 to December 2023.

Table A9. RTP and the presence of other investors

This table reports Fama–MacBeth regressions of the one-month-ahead return gap (in percent) on RTP and its interactions with indicators for the presence of other investors and for their absorption of retail buying, following the specification of Appendix Table A13. High short-term institution trading proportion indicates an above-median trading proportion of short-term institutions within the month, constructed from the trade records in the same way as RTP. High foreign ownership indicates above-median foreign ownership within the month. High foreign trading proportion indicates an above-median trading proportion of foreign investors within the month, constructed in the same way. High short-term and foreign absorption share indicates an above-median share of the stock's retail buy volume during the first thirty minutes of continuous trading executed against short-term institutional or foreign sellers in that month. Columns 2 to 5 enter one interaction at a time, column 6 enters the three presence interactions jointly, and column 7 enters the absorption interaction together with the two trading-proportion interactions. The indicators themselves are included in the regressions but not tabulated, and all specifications include the full control set of Table 7 and industry fixed effects. RTP and all controls are winsorized at the 1st and 99th percentiles and standardized within month. Months is the number of monthly cross-sections. Newey–West standard errors with six lags are in parentheses; ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample is the larger half of stocks by beginning-of-year market capitalization from January 2009 to December 2023.

	Return gap						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RTP	2.35***	2.83***	2.80***	2.97***	2.81***	3.18***	3.06***
	(0.18)	(0.33)	(0.26)	(0.33)	(0.32)	(0.38)	(0.40)
RTP × High indicator for:							
Short-term institution trading proportion		-0.62**				-0.23	0.06
		(0.26)				(0.27)	(0.26)
Foreign ownership			-0.61***			-0.39***	
			(0.17)			(0.15)	
Foreign trading proportion				-0.76***		-0.42*	-0.25
				(0.26)		(0.22)	(0.22)
Short-term and foreign absorption share					-0.79***		-0.80***
					(0.24)		(0.22)
Controls	✓	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓	✓
Months	180	180	180	180	178	180	178
N	159,472	159,472	159,472	159,472	158,051	159,472	158,051
Adjusted R^2 (%)	16.02	16.14	16.12	16.11	16.08	16.20	16.13

Appendix F Additional figures and tables

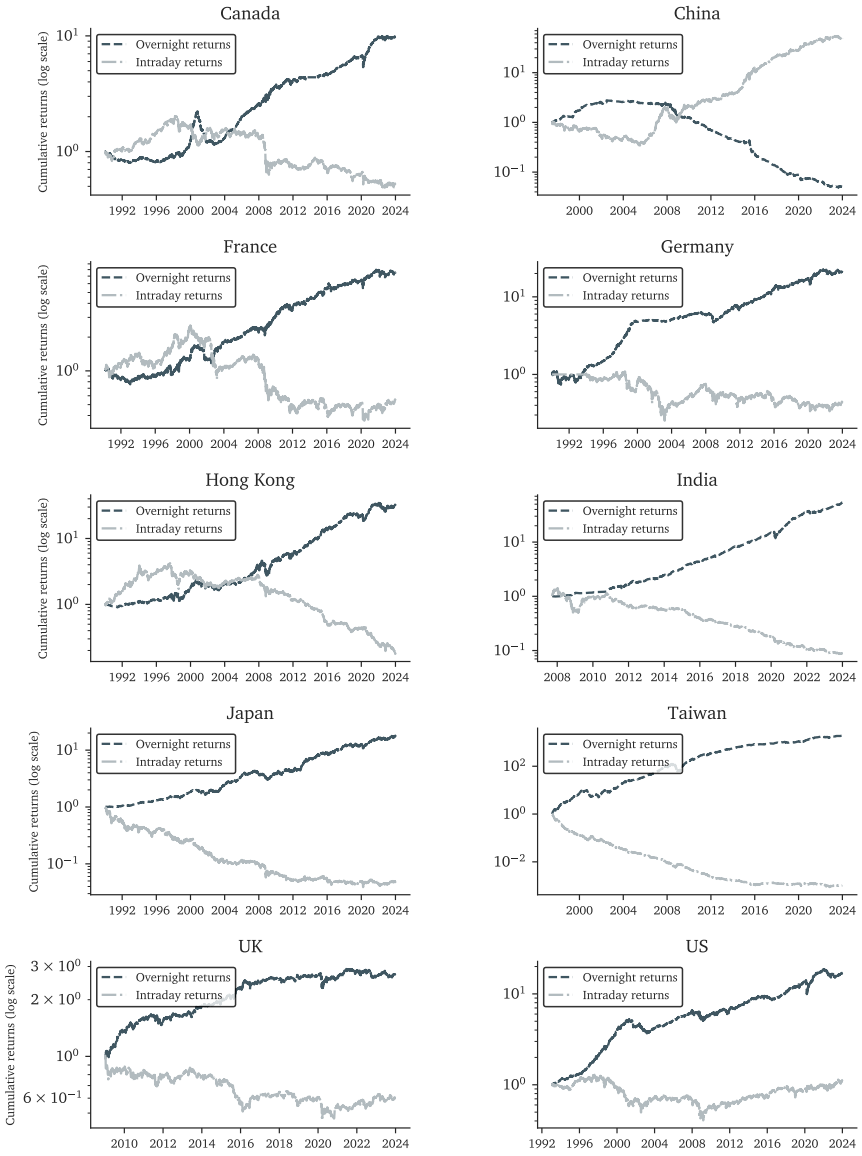
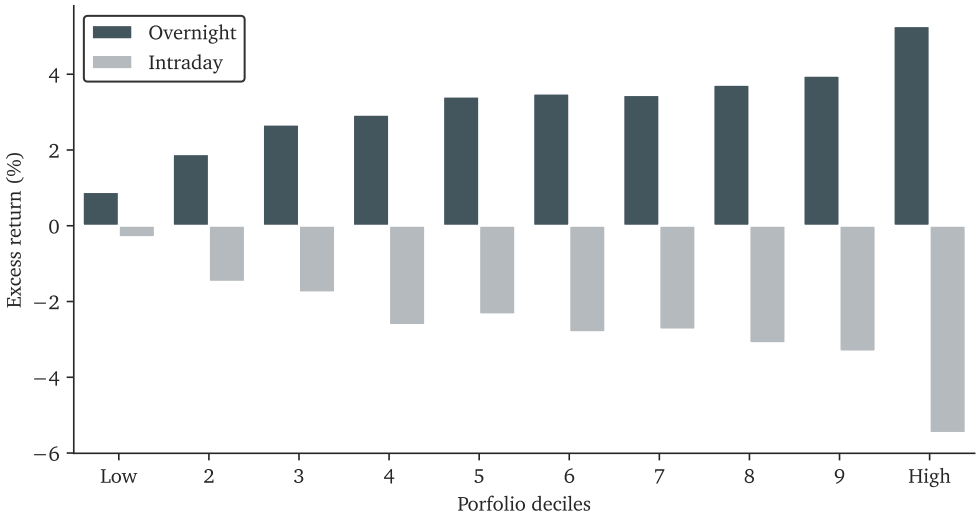
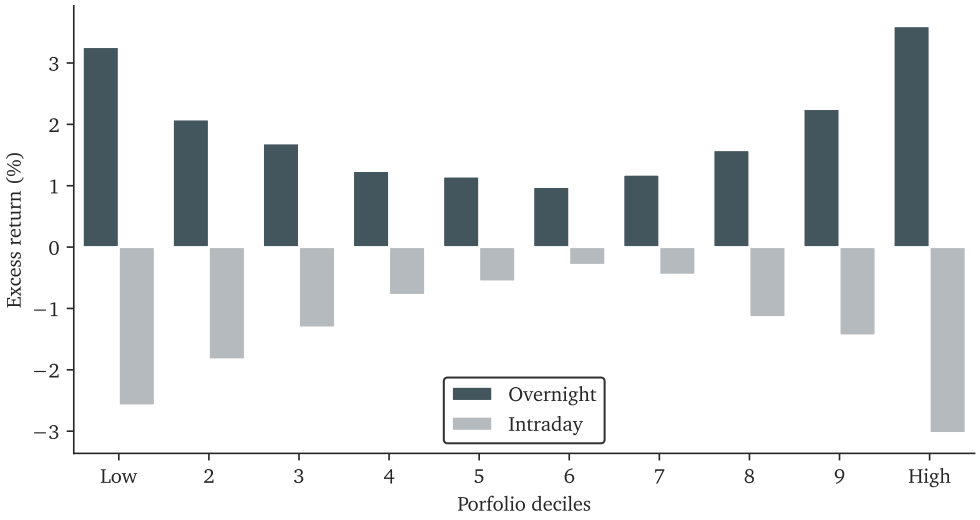


Figure A5. Cumulative overnight and intraday returns across countries

This figure plots cumulative overnight and intraday returns for major equity market indices from January 2010 to December 2023 on a log scale. For the U.S. and U.K., we use liquid exchange-traded funds (ETFs) to compute intraday returns. For other markets, we use the underlying index tickers directly, as corresponding ETFs were either introduced later or have insufficient liquidity.



(a) Portfolios sorted by RTP



(b) Portfolios sorted by RNBO minus RNBC

Figure A6. Average overnight and intraday returns of sorted portfolios

This figure plots the monthly average overnight and intraday returns of portfolios sorted by variables based on retail trading. Each month, stocks are sorted into deciles based on the sorting variable. Panel (a) shows the results sorted by retail trading proportion (RTP). Panel (b) shows the results sorted by the difference between the retail net buy at market open and close (RNBO minus RNBC). RNBO and RNBC are retail net buy, scaled by shares outstanding and summed within the month, in the first hour of trading and in the last hour of trading. Portfolios are value-weighted and rebalanced monthly. The sample period is January 2009 to December 2023.

Table A10. RTP and net buy flows by retail trading medium

This table repeats the Fama–MacBeth regressions of Table 9 with the retail net buy flow decomposed by trading channel: MTS (mobile trading systems), HTS (home trading systems), and Other (branch, telephone, and other channels), identified from the order-medium code of the KRX trade records. The three channel flows sum to the retail flow of Table 9 before winsorization, so the channel coefficients add up to the retail coefficients only approximately. The MTS columns coincide with its Mobile trader columns. The dependent variables are net buy flows (buy minus sell volume, scaled by shares outstanding) measured in basis points, using trades executed in the first hour of trading in columns 1–3 and in the last hour of trading in columns 4–6, averaged over the next 1, 5, and 20 days in Panels A, B, and C. RTP is measured on day t, and control variables are the same as those in Table 7. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation, and the dependent flows are winsorized at the 1% and 99% levels within day. Coefficients are scaled by 100 as in Table 9. Newey–West standard errors are reported in parentheses, computed with 20 lags. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Net buy at open			Net buy at close		
	(1) MTS	(2) HTS	(3) Other	(4) MTS	(5) HTS	(6) Other
Panel A: Flows on the next day						
RTP	21.51***	-3.61***	-9.74***	-10.40***	17.29***	6.48***
	(1.31)	(0.53)	(0.39)	(0.70)	(1.05)	(0.41)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709
Adjusted R^2 (%)	5.02	2.11	2.04	4.23	1.68	1.05
Panel B: Average flows over the next 5 days						
RTP	21.75***	-1.38**	-12.10***	-11.81***	17.99***	5.53***
	(1.27)	(0.55)	(0.45)	(0.79)	(0.99)	(0.38)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709
Adjusted R^2 (%)	6.76	2.42	3.93	8.50	2.75	1.53
Panel C: Average flows over the next 20 days						
RTP	22.05***	-1.65***	-12.67***	-13.24***	17.42***	4.67***
	(1.25)	(0.59)	(0.44)	(0.85)	(0.87)	(0.34)
Controls	✓	✓	✓	✓	✓	✓
N	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709	3,408,709
Adjusted R^2 (%)	9.55	3.31	8.07	12.13	4.01	2.19

Table A11. RTP and net buy flows, full coefficient estimates

This table reports the full coefficient estimates of the next-day specification in Panel A of Table 9, which displays the RTP coefficient only. The dependent variables are investor net buy flows (buy minus sell volume, scaled by shares outstanding) on the day following day t, measured in basis points. Columns 1–3 use trades executed in the first hour of trading and columns 4–6 trades executed in the last hour of trading. The column headers indicate the investor group as in Table 9. RTP and all controls are measured on day t, winsorized at the 1% and 99% levels, and standardized cross-sectionally to have zero mean and unit standard deviation. The dependent flows are winsorized at the 1% and 99% levels within day. Coefficients are scaled by 100 for readability. Newey–West standard errors are reported in parentheses, computed with 20 lags. The sample period is January 2009 to December 2023 for retail and mobile trades, and January 2015 to December 2018 for day trades. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Net buy at open			Net buy at close		
	(1) Retail	(2) Mobile	(3) Day trader	(4) Retail	(5) Mobile	(6) Day trader
RTP	8.10***	21.51***	3.12***	13.04***	-10.40***	-1.65***
	(1.21)	(1.31)	(0.26)	(1.86)	(0.70)	(0.14)
Overnight return	9.87***	16.83***	0.97	-2.98***	-6.39***	-0.46**
	(1.19)	(1.38)	(0.59)	(0.50)	(0.53)	(0.19)
Intraday return	70.28***	62.72***	1.44***	-6.82***	-7.86***	-0.76***
	(4.39)	(4.30)	(0.30)	(1.02)	(0.62)	(0.08)
Log market equity	2.95**	7.99***	-1.05***	7.55***	-0.83	1.16***
	(1.16)	(0.88)	(0.10)	(1.03)	(0.54)	(0.07)
Market beta	0.82	-0.80	0.23	-1.09	0.65	-0.27***
	(0.80)	(0.79)	(0.14)	(0.84)	(0.72)	(0.07)
Book-to-market	0.46	0.19	-0.27***	6.78***	1.20***	0.44***
	(0.66)	(0.53)	(0.10)	(0.62)	(0.41)	(0.08)
Amihud illiquidity	-4.91***	-7.28***	-1.83***	10.85***	8.20***	1.34***
	(0.58)	(0.58)	(0.15)	(0.55)	(0.53)	(0.10)
Foreign ownership	2.67***	2.25***	0.80***	6.29***	1.57***	-0.41***
	(0.53)	(0.40)	(0.06)	(0.49)	(0.24)	(0.04)
Past 120-day return	-2.08	11.88***	1.94***	-29.08***	-17.24***	-1.10***
	(1.42)	(1.26)	(0.17)	(0.88)	(0.90)	(0.12)
Idiosyncratic volatility	27.75***	28.43***	4.51***	-1.71	-14.16***	-3.12***
	(1.81)	(1.78)	(0.35)	(1.40)	(1.15)	(0.22)
Maximum daily return	4.91***	21.06***	3.37***	-17.50***	-24.56***	-1.38***
	(1.82)	(2.18)	(0.34)	(1.17)	(1.64)	(0.20)
N	3,408,709	3,408,709	896,999	3,408,709	3,408,709	896,999
Adjusted R^2 (%)	5.69	5.02	5.38	3.48	4.23	4.33

Table A12. Robustness to alternative mechanisms

This table presents results from Fama–MacBeth regressions of one-month-ahead overnight-intraday return-gap on retail trading proportion (RTP) and other variables proposed in prior studies to explain overnight and intraday returns. Returns are expressed in percentage points. All columns include the control variables of Table 7. Detailed definitions of all variables are provided in Appendix Table A1. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Newey–West standard errors are reported in parentheses, computed with six lags. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return gap					
	(1)	(2)	(3)	(4)	(5)	(6)
RTP	2.43***	2.40***	2.09***	2.17***	2.16***	2.14***
	(0.18)	(0.19)	(0.16)	(0.16)	(0.15)	(0.15)
Idiosyncratic volatility	2.31***	2.32***	2.18***	2.15***	2.15***	2.09***
	(0.19)	(0.20)	(0.19)	(0.19)	(0.19)	(0.19)
Abnormal volume		-0.10	-0.03	-0.05	-0.04	-0.04
		(0.12)	(0.12)	(0.12)	(0.12)	(0.12)
Overnight systematic variance			0.99***	0.98***	1.20***	1.21***
			(0.14)	(0.14)	(0.25)	(0.25)
Retail order imbalance (open)				0.27***	0.28***	0.28***
				(0.05)	(0.05)	(0.05)
Overnight beta					-0.25	-0.26
					(0.27)	(0.27)
Abnormal night reversal						-0.22***
						(0.05)
N	169,961	167,085	163,985	163,985	163,985	163,985
Adjusted R^2 (%)	13.74	14.01	14.32	14.38	14.49	14.52

Table A13. Heterogeneous effects of RTP across subsamples

This table presents results from Fama–MacBeth regressions testing whether RTP's predictive power varies with proxies for alternative mechanisms. The dependent variable is one-month-ahead overnight-intraday return gap. Returns are expressed in percentage points. The parameter of interest is the interaction between RTP and an indicator equal to one when the mechanism variable exceeds its cross-sectional median. Columns 1–6 enter one mechanism variable at a time, and column 7 enters all six jointly. The indicators for overnight systematic variance and overnight beta nearly coincide, so their interactions are not separately identified in column 7. All columns include the control variables of Table 7, and Appendix Table A1 defines all variables. All independent variables are winsorized at the 1% and 99% levels and standardized cross-sectionally to have zero mean and unit standard deviation. Newey–West standard errors are reported in parentheses, computed with six lags. The sample period is January 2009 to December 2023. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Return gap						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RTP	2.14***	2.35***	2.25***	2.17***	2.29***	2.47***	1.74***
	(0.17)	(0.20)	(0.15)	(0.22)	(0.15)	(0.20)	(0.18)
RTP × High indicator for:							
Idiosyncratic volatility	0.79***						0.79***
	(0.12)						(0.12)
Abnormal volume		0.11					-0.09
		(0.10)					(0.09)
Overnight systematic variance			0.07				2.61
			(0.12)				(1.89)
Retail order imbalance (open)				0.47***			0.54***
				(0.13)			(0.13)
Overnight beta					0.06		-2.67
					(0.13)		(1.90)
Abnormal night reversal						-0.10	-0.05
						(0.08)	(0.08)
N	169,961	167,085	166,787	169,961	169,171	168,660	163,985
Adjusted R^2 (%)	13.87	13.87	13.82	13.84	13.92	13.77	14.12

Table A14. Summary statistics: Additional variables

This table reports summary statistics for the remaining regression variables of the paper, defined in Appendix Table A1. Panel A covers the five mechanism variables of Tables 11 and 12, computed on the raw monthly values before the within-month winsorization and standardization applied in the regressions. Overnight systematic variance is in squared percent and retail order imbalance at the open is in percent of daily volume, while the remaining variables are unitless. Panel B covers the stock-day variables of the flow-prediction regressions of Table 9: the same-day net buy flows whose 1-, 5-, and 20-day leads serve as dependent variables (in basis points of shares outstanding), daily RTP and its trailing 20-day mean, and the daily controls. All are computed on the raw values before the within-day winsorization and standardization applied in the regressions. Panel A uses the stock-month panel of Table 1 and Panel B its stock-day counterpart over the same universe (January 2009 to December 2023, with day-trade flows from the brokerage data for January 2015 to December 2018), with per-variable counts reflecting availability.

	N	Mean	SD	Min	25th	50th	75th	Max
Panel A: Stock-month variables								
Abnormal volume	167,085	-0.03	0.52	-7.12	-0.10	-0.02	0.01	14.55
Overnight systematic variance	166,787	0.52	0.63	0.00	0.11	0.29	0.68	6.86
Retail order imbalance (open)	169,961	-1.10	1.36	-23.87	-1.84	-0.98	-0.28	27.67
Overnight beta	169,171	0.87	0.39	-2.97	0.59	0.87	1.14	3.95
Abnormal night reversal	168,660	1.00	0.35	0.00	0.77	0.99	1.22	4.68
Panel B: Stock-day variables of Table 9								
Retail net buy at open (bps)	3,529,166	0.43	9.43	-1,471.27	-0.79	0.02	1.13	2,921.28
Retail net buy at close (bps)	3,529,166	-0.50	5.40	-437.96	-1.20	-0.05	0.56	978.92
Mobile net buy at open (bps)	3,529,166	0.64	11.39	-958.77	-0.54	0.03	0.98	2,931.08
Mobile net buy at close (bps)	3,529,166	-0.46	5.52	-793.65	-0.88	-0.04	0.49	555.64
Day-trade net buy at open (bps)	927,427	0.10	1.14	-158.17	0.00	0.00	0.04	122.77
Day-trade net buy at close (bps)	927,427	-0.07	0.76	-115.20	-0.01	0.00	0.00	108.73
RTP (%)	3,529,166	72.88	21.53	0.00	59.63	78.57	90.16	100.00
RTP, past 20-day mean (%)	3,529,166	72.94	20.00	3.95	61.24	78.48	88.43	100.00
Overnight return (%)	3,529,166	0.11	1.46	-30.00	-0.44	0.00	0.65	30.01
Intraday return (%)	3,529,166	-0.07	2.83	-39.20	-1.46	-0.14	1.15	66.67
Past 120-day return (%)	3,509,866	5.41	41.17	-97.33	-16.54	-2.15	16.87	2,033.46
Log market equity	3,529,131	5.89	1.31	1.92	4.98	5.57	6.49	13.21
Market beta	3,509,866	0.88	0.50	-3.58	0.53	0.86	1.21	5.40
Book-to-market ratio	3,529,131	0.99	0.90	0.00	0.38	0.74	1.32	55.52
Amihud illiquidity	3,408,709	0.06	0.83	0.00	0.00	0.01	0.03	152.32
Foreign ownership (%)	3,529,131	10.27	13.36	0.00	1.56	5.06	13.61	100.00
Idiosyncratic volatility (%)	3,529,166	2.13	1.24	0.17	1.34	1.83	2.54	18.90
Maximum daily return (%)	3,529,166	6.05	4.54	-0.12	3.21	4.82	7.27	30.00